# BOOK 1
# PHYSICAL QUANTITIES & UNITS

**1 Physical quantities and units**

**1.1 Physical quantities**

Candidates should be able to:

1 understand that all physical quantities consist of a numerical magnitude and a unit

2 make reasonable estimates of physical quantities included within the syllabus

**1.2 SI units**

Candidates should be able to:

1 recall the following SI base quantities and their units: mass (kg), length (m), time (s), current (A), temperature (K)

2 express derived units as products or quotients of the SI base units and use the derived units for quantities listed in this syllabus as appropriate

3 use SI base units to check the homogeneity of physical equations

4 recall and use the following prefixes and their symbols to indicate decimal submultiples or multiples of both base and derived units: pico (p), nano (n), micro (μ), milli (m), centi (c), deci (d), kilo (k), mega (M), giga (G), tera (T)

**1.4 Scalars and vectors**

Candidates should be able to:

1 understand the difference between scalar and vector quantities and give examples of scalar and vector quantities included in the syllabus

2 add and subtract coplanar vectors

3 represent a vector as two perpendicular components

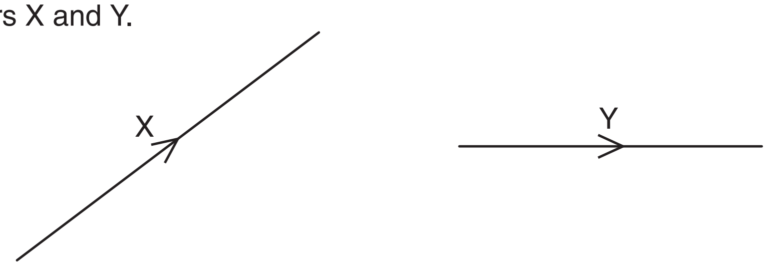
1 **9702/1/M/J/02/Q1**

Which of the following pairs of units are both SI base units?

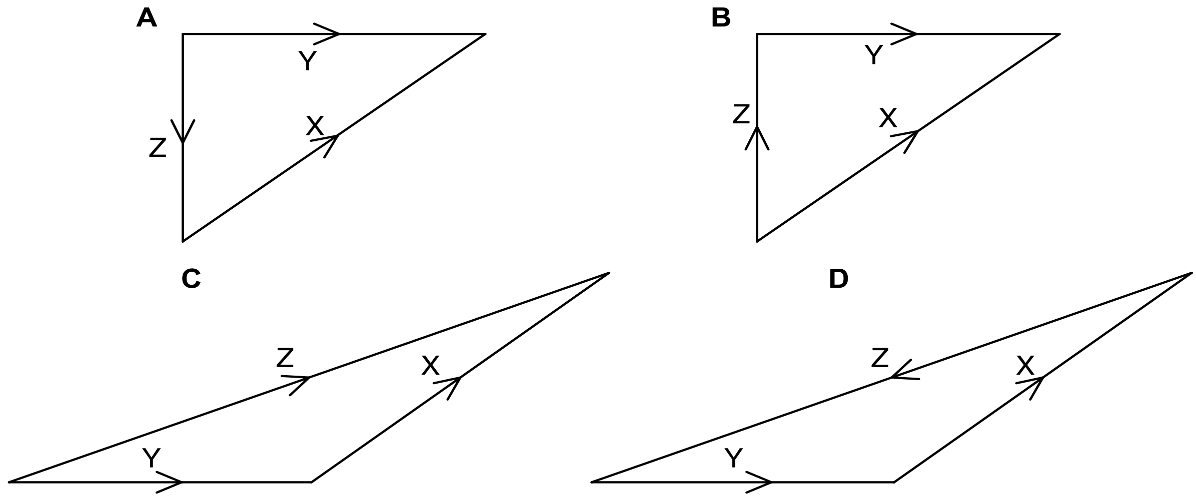
**A** ampere, degree celsius
**B** ampere, kelvin
**C** coulomb, degree celsius
**D** coulomb, kelvin

2 **9702/1/M/J/02/Q2***

The diagram shows two vectors X and Y.




In which vector triangle does the vector Z show the magnitude and direction of vector X – Y?




3 **9702/1/O/N/02/Q1**

The prefix 'centi' indicates x $10^{-2}$. That is, 1 centimetre is equal to $1 \times 10^{-2}$ metre.

Which line in the table correctly indicates the prefixes micro, nano and pico?

| | $\times 10^{-12}$ | $\times 10^{-9}$ | $\times 10^{-6}$ |
|---|---|---|---|
| **A** | nano | micro | pico |
| **B** | micro | pico | nano |
| **C** | pico | nano | micro |
| **D** | pico | micro | nano |

4 **9702/01/M/J/03/Q1**

Which of the following is a scalar quantity?

**A** acceleration **B** mass **C** momentum **D** velocity

**5** **9702/01/M/J/03/Q2**

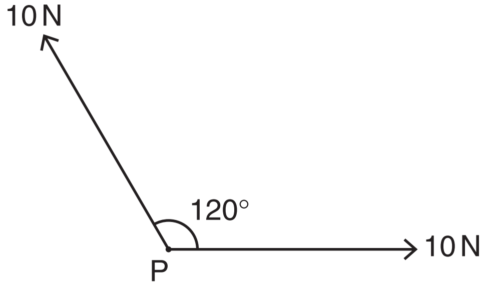
The unit of work, the joule, may be defined as the work done when the point of application of a force of 1 newton is moved a distance of 1 metre in the direction of the force.

Express the joule in terms of the base units of mass, length and time, the kg, m and s.

**A** $kg\,m^{-1}\,s^{2}$ **B** $kg\,m^{2}\,s^{-2}$ **C** $kg\,m^{2}\,s^{-1}$ **D** $kg\,s^{-2}$

**6** **9702/01/M/J/03/Q3**

Two forces, each of 10 N, act at a point P as shown in the diagram. The angle between the directions of the forces is 120°.




What is the magnitude of the resultant force?

**A** 5 N **B** 10 N **C** 17 N **D** 20 N

**7** **9702/01/O/N/03/Q1**

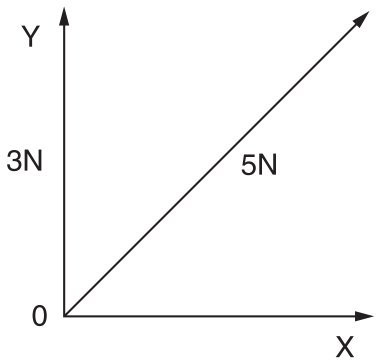
A student measures a current as 0.5 A.

Which of the following correctly expresses this result?

**A** 50 mA **B** 50 MA **C** 500 mA **D** 500 MA

**8** **9702/01/O/N/03/Q2**

A force of 5 N may be represented by two perpendicular components OY and OX as shown in the diagram, which is **not** drawn to scale.




OY is of magnitude 3 N.

What is the magnitude of OX?

**A** 2 N **B** 3 N **C** 4 N **D** 5 N

**9** **9702/01/O/N/03/Q3**

The momentum of an object of mass $m$ is $p$.

Which quantity has the same base units as $\frac{p^2}{m}$?

**A** energy **B** force **C** power **D** velocity

**10** **9702/11/M/J/15/Q2***

Which pair contains one vector and one scalar quantity?

**A** displacement : acceleration

**B** force : kinetic energy

**C** momentum : velocity

**D** power : speed

**11** **9702/01/M/J/04/Q2**

Which of the following could be measured in the same units as force?

**A** energy / distance

**B** energy x distance

**C** energy / time

**D** momentum x distance

**12 9702/01/M/J/04/Q3**

The notation μs is used as an abbreviation for a certain unit of time.

What is the name and value of this unit?

| | name | value |
|---|---|---|
| **A** | microsecond | $10^{-6}$ s |
| **B** | microsecond | $10^{-3}$ s |
| **C** | millisecond | $10^{-6}$ s |
| **D** | millisecond | $10^{-3}$ s |

**13 9702/01/O/N/04/Q1**

Which line of the table gives values that are equal to a time of 1 ps (one picosecond) and a distance of 1 Gm (one gigametre)?

| | time of 1 ps | distance of 1 Gm |
|---|---|---|
| **A** | $10^{-9}$ s | $10^{9}$ m |
| **B** | $10^{-9}$ s | $10^{12}$ m |
| **C** | $10^{-12}$ s | $10^{9}$ m |
| **D** | $10^{-12}$ s | $10^{12}$ m |

**14 9702/01/O/N/04/Q2**

Which of the following definitions is correct and uses only quantities rather than units?

**A** Density is mass per cubic metre.

**B** Potential difference is energy per unit current.

**C** Pressure is force per unit area.

**D** Speed is distance travelled per second.

**15 9702/01/O/N/04/Q3**

When a beam of light is incident on a surface, it delivers energy to the surface. The intensity of the beam is defined as the energy delivered per unit area per unit time.

What is the unit of intensity, expressed in SI base units?

**A** $\mathrm{kg\,m^{-2}\,s^{-1}}$ **B** $\mathrm{kg\,m^{2}\,s^{-3}}$ **C** $\mathrm{kg\,s^{-2}}$ **D** $\mathrm{kg\,s^{-3}}$

**16 9702/01/M/J/05/Q1***

Decimal sub-multiples and multiples of units are indicated using a prefix to the unit. For example, the prefix milli (m) represents $10^{-3}$.

Which of the following gives the sub-multiples or multiples represented by pico (p) and giga (G)?

| | pico (p) | giga (G) |
|---|---|---|
| **A** | $10^{-9}$ | $10^{9}$ |
| **B** | $10^{-9}$ | $10^{12}$ |
| **C** | $10^{-12}$ | $10^{9}$ |
| **D** | $10^{-12}$ | $10^{12}$ |

**17 9702/01/M/J/05/Q2***

A metal sphere of radius *r* is dropped into a tank of water. As it sinks at speed *v*, it experiences a drag force *F* given by *F* = *krv*, where *k* is a constant.

What are the SI base units of *k*?

**A** $\mathrm{kg\,m^{2}\,s^{-1}}$ **B** $\mathrm{kg\,m^{-2}\,s^{-2}}$ **C** $\mathrm{kg\,m^{-1}\,s^{-1}}$ **D** $\mathrm{kg\,m\,s^{-2}}$

**18 9702/11/O/N/10/Q2***

An Olympic athlete of mass 80 kg competes in a 100 m race.

What is the best estimate of his mean kinetic energy during the race?

**A** $4 \times 10^{2}$ J **B** $4 \times 10^{3}$ J **C** $4 \times 10^{4}$ J **D** $4 \times 10^{5}$ J

**19 9702/01/O/N/05/Q1**

Which pair of units are both SI base units?

**A** ampere, degree celsius
**B** ampere, kelvin
**C** coulomb, degree celsius
**D** coulomb, kelvin

**20 9702/01/O/N/05/Q2**

The prefix 'centi' indicates x $10^{-2}$.

Which line in the table correctly indicates the prefixes micro, nano and pico?

| | x $10^{-12}$ | x $10^{-9}$ | x $10^{-6}$ |
|---|---|---|---|
| **A** | nano | micro | pico |
| **B** | nano | pico | micro |
| **C** | pico | nano | micro |
| **D** | pico | micro | nano |

**21 9702/01/O/N/05/Q3**

Which expression involving base units is equivalent to the volt?

**A** $\mathrm{kg\,m^{2}\,s^{-1}\,A^{-1}}$ **B** $\mathrm{kg\,m\,s^{-2}\,A}$ **C** $\mathrm{kg\,m^{2}\,s^{-1}\,A}$ **D** $\mathrm{kg\,m^{2}\,s^{-3}\,A^{-1}}$

**22 9702/12/M/J/13/Q1****

Which pair includes a vector quantity and a scalar quantity?

**A** displacement; acceleration
**B** force; kinetic energy
**C** power; speed
**D** work; potential energy

**23 9702/12/M/J/12/Q2**

For which quantity is the magnitude a reasonable estimate?

**A** frequency of a radio wave 500 pHz
**B** mass of an atom 500 µg
**C** the Young modulus of a metal 500 kPa
**D** wavelength of green light 500 nm

**24 9702/01/M/J/06/Q3***

The following physical quantities can be either positive or negative.

*s* : displacement of a particle along a straight line

*q* : electric charge

*θ* : temperature on the Celsius scale

*V* : readings on a digital voltmeter

Which of these quantities are vectors?

**A** *s*, *θ*, *q*, *V* **B** *s*, *q*, *V* **C** *θ*, *V* **D** *s* only

**25** **9702/01/O/N/06/Q1**

Which product-pair of metric prefixes has the greatest magnitude?

**A** pico × mega **B** nano × kilo **C** micro × giga **D** milli × tera

**26** **9702/01/O/N/06/Q2**

In the expressions below

*a* is acceleration, *F* is force, *m* is mass, *t* is time, *v* is velocity.

Which expression represents energy?

**A** $Ft$ **B** $Fvt$ **C** $\frac{2mv}{t}$ **D** $\frac{at^2}{2}$

**27** **9702/01/O/N/06/Q3**

Which row of the table shows a physical quantity and its correct unit?

| | physical quantity | unit |
|---|---|---|
| **A** | electric field strength | $\text{kg m s}^{-2}\,\text{C}^{-1}$ |
| **B** | specific heat capacity | $\text{kg}^{-1}\,\text{m}^2\,\text{s}^{-2}\,\text{K}^{-1}$ |
| **C** | tensile strain | $\text{kg m}^{-1}\,\text{s}^{-2}$ |
| **D** | the Young modulus | $\text{kg m}^{-1}\,\text{s}^{-3}$ |

**28** **9702/01/M/J/07/Q1**

Which is a pair of SI base units?

| | | |
|---|---|---|
| **A** | ampere | joule |
| **B** | coulomb | second |
| **C** | kilogram | kelvin |
| **D** | metre | newton |

**29** **9702/01/M/J/07/Q2**

What is the ratio $\frac{1\,\mu\text{m}}{1\,\text{Gm}}$?

**A** $10^{-3}$ **B** $10^{-9}$ **C** $10^{-12}$ **D** $10^{-15}$

**30** **9702/01/M/J/07/Q3**

Which formula could be correct for the speed $v$ of ocean waves in terms of the density $\rho$ of sea-water, the acceleration of free fall $g$, the depth $h$ of the ocean and the wavelength $\lambda$?

**A** $v = \sqrt{g\lambda}$ **B** $v = \sqrt{\frac{g}{h}}$ **C** $v = \sqrt{\rho g h}$ **D** $v = \sqrt{\frac{g}{\rho}}$

**31** **9702/01/O/N/07/Q1**

The equation relating pressure and density is $p = \rho g h$.

How can both sides of this equation be written in terms of base units?

**A** $[\text{N m}^{-1}] = [\text{kg m}^{-3}]\,[\text{m s}^{-1}]\,[\text{m}]$

**B** $[\text{N m}^{-2}] = [\text{kg m}^{-3}]\,[\text{m s}^{-2}]\,[\text{m}]$

**C** $[\text{kg m}^{-1}\,\text{s}^{-2}] = [\text{kg m}^{-3}]\,[\text{m s}^{-2}]\,[\text{m}]$

**D** $[\text{kg m}^{-1}\,\text{s}^{-1}] = [\text{kg m}^{-1}]\,[\text{m s}^{-2}]\,[\text{m}]$

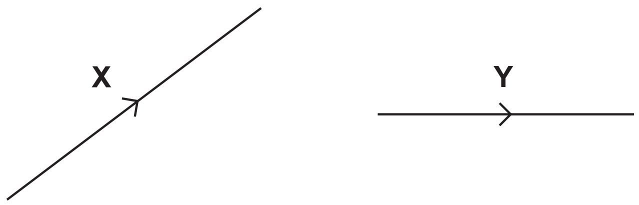
**32** **9702/01/O/N/07/Q2**

What is a reasonable estimate of the diameter of an alpha particle?

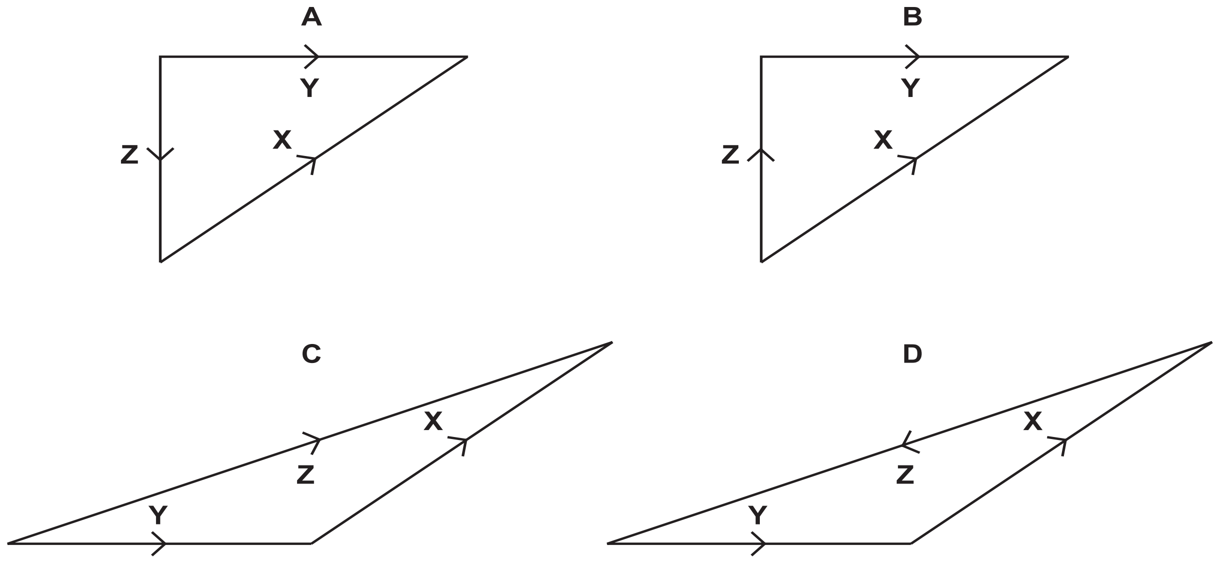
**A** $10^{-15}$ m **B** $10^{-12}$ m **C** $10^{-9}$ m **D** $10^{-6}$ m

**33 9702/01/O/N/07/Q3**

The diagram shows two vectors **X** and **Y**.




In which vector triangle does the vector **Z** show the magnitude and direction of vector **X–Y**?




**34 9702/01/M/J/08/Q1**

Five energies are listed.

5 kJ 5 mJ 5 MJ 5 nJ

Starting with the smallest first, what is the order of increasing magnitude of these energies?

**A** 5 kJ → 5 mJ → 5 MJ → 5 nJ

**B** 5 nJ → 5 kJ → 5 MJ → 5 mJ

**C** 5 nJ → 5 mJ → 5 kJ → 5 MJ

**D** 5 mJ → 5 nJ → 5 kJ → 5 MJ

**35 9702/01/M/J/08/Q2**

Which of the following correctly expresses the volt in terms of SI base units?

**A** $\mathrm{A\,\Omega}$ **B** $\mathrm{W\,A^{-1}}$ **C** $\mathrm{kg\,m^2\,s^{-1}\,A^{-1}}$ **D** $\mathrm{kg\,m^2\,s^{-3}\,A^{-1}}$

**36 9702/13/M/J/12/Q1***

What is a reasonable estimate of the average kinetic energy of an athlete during a 100 m race that takes 10 s?

**A** 40 J **B** 400 J **C** 4000 J **D** 40 000 J

**37 9702/01/O/N/08/Q1**

A laser emits light of wavelength 600 nm.

What is the distance, expressed as a number of wavelengths, travelled by the light in one second?

**A** $5 \times 10^{8}$ **B** $5 \times 10^{11}$ **C** $5 \times 10^{14}$ **D** $5 \times 10^{17}$

**38 9702/01/O/N/08/Q2**

At temperatures close to 0 K, the specific heat capacity $c$ of a particular solid is given by $c = bT^3$, where $T$ is the thermodynamic temperature and $b$ is a constant characteristic of the solid.

What are the units of constant $b$, expressed in SI base units?

**A** $m^2 s^{-2} K^{-3}$ **B** $m^2 s^{-2} K^{-4}$ **C** $kg\, m^2 s^{-2} K^{-3}$ **D** $kg\, m^2 s^{-2} K^{-4}$

**39 9702/01/O/N/08/Q3**

The table shows the $x$-component and $y$-component of four force vectors.

Which force vector has the largest magnitude?

| | $x$-component / N | $y$-component / N |
|---|---|---|
| **A** | 2 | 9 |
| **B** | 3 | 8 |
| **C** | 4 | 7 |
| **D** | 5 | 6 |

**40 9702/01/M/J/09/Q1**

Which statement, involving multiples and sub-multiples of the base unit metre (m), is correct?

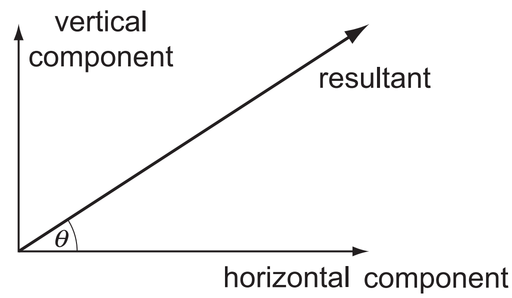
**A** 1 pm = $10^{-9}$ m **B** 1 nm = $10^{-6}$ m **C** 1 mm = $10^{6}$ μm **D** 1 km = $10^{6}$ mm

**41 9702/01/M/J/09/Q2**

The diagram shows a resultant force and its horizontal and vertical components.

The horizontal component is 20.0 N and $\theta$ = 30°

. What is the vertical component?




**A** 8.7 N **B** 10.0 N

**C** 11.5 N **D** 17.3 N

**42 9702/12/O/N/09/Q1**

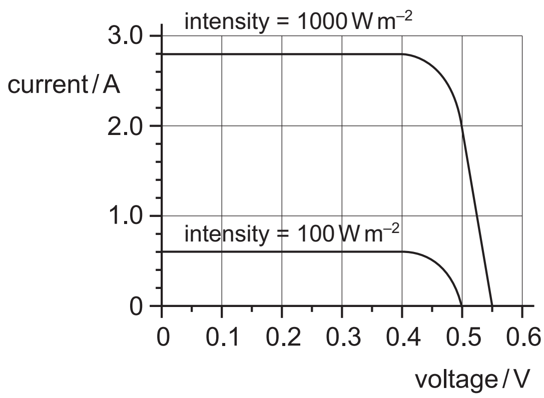
The drag force $F$ acting on a moving sphere obeys an equation of the form $F = kAv^2$, where $A$ represents the sphere's frontal area and $v$ represents its speed.

What are the base units of the constant $k$?

**A** $kg\, m^5 s^{-4}$ **B** $kg\, m^{-2} s^{-1}$ **C** $kg\, m^{-3}$ **D** $kg\, m^{-4} s^2$

**43 9702/12/O/N/09/Q2**

The graph shows two current-voltage calibration curves for a solar cell exposed to different light intensities.




At zero voltage, what is the ratio

$$\frac{\text{current at } 1000\,\text{W m}^{-2}}{\text{current at } 100\,\text{W m}^{-2}}\ ?$$

**A** 1.1 **B** 4.7

**C** 8.0 **D** 10

**44** **9702/12/O/N/09/Q40**

The table contains some quantities, together with their symbols and units.

| quantity | symbol | unit |
|---|---|---|
| gravitational field strength | $g$ | $\mathrm{N\,kg^{-1}}$ |
| density of liquid | $\rho$ | $\mathrm{kg\,m^{-3}}$ |
| vertical height | $h$ | m |
| volume of part of liquid | $V$ | $\mathrm{m^3}$ |

Which expression has the units of energy?

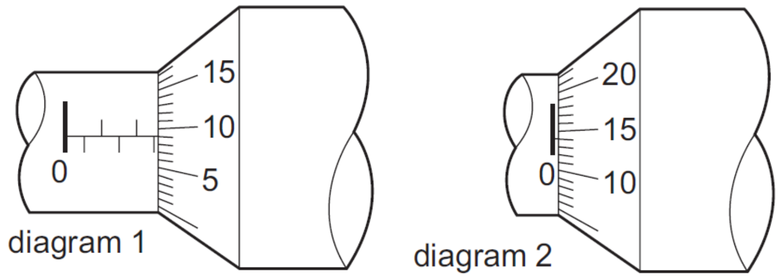
**A** $g\rho hV$ **B** $\dfrac{\rho hV}{g}$

**C** $\dfrac{\rho g}{hV}$ **D** $\rho g^2 h$

**45** **9702/12/M/J/10/Q1**

A micrometer screw gauge is used to measure the diameter of a copper wire.

The reading with the wire in position is shown in diagram 1. The wire is removed and the jaws of the micrometer are closed. The new reading is shown in diagram 2.




What is the diameter of the wire?

**A** 1.90 mm **B** 2.45 mm

**C** 2.59 mm **D** 2.73 mm

**46** **9702/12/M/J/10/Q2**

The SI unit for potential difference (the volt) is given, in base units, by

**A** $\mathrm{kg\,m\,A^{-1}\,s^{-3}}$. **B** $\mathrm{m^2\,A^{-1}\,s^{-2}}$. **C** $\mathrm{kg\,m^2\,s^{-2}}$. **D** $\mathrm{kg\,m^2\,A^{-1}\,s^{-3}}$.

**47** **9702/12/M/J/10/Q7**

The product of pressure and volume has the same SI base units as

**A** energy. **B** force. **C** $\dfrac{\text{force}}{\text{area}}$. **D** $\dfrac{\text{force}}{\text{length}}$.

**48** **9702/11/O/N/10/Q1**

A signal has a frequency of 2.0 MHz.

What is the period of the signal?

**A** 2 µs **B** 5 µs **C** 200 ns **D** 500 ns

**49** **9702/01/M/J/05/Q2**

A metal sphere of radius $r$ is dropped into a tank of water. As it sinks at speed $v$, it experiences a drag force $F$ given by $F = krv$, where $k$ is a constant.

What are the SI base units of $k$?

**A** $\mathrm{kg\,m^2\,s^{-1}}$ **B** $\mathrm{kg\,m^{-2}\,s^{-2}}$ **C** $\mathrm{kg\,m^{-1}\,s^{-1}}$ **D** $\mathrm{kg\,m\,s^{-2}}$

**50** **9702/11/O/N/10/Q3**

Which physical quantity would result from a calculation in which a potential difference is multiplied by an electric charge?

**A** electric current **B** electric energy **C** electric field strength **D** electric power

**51** **9702/12/O/N/10/Q1**

Which row shows a base quantity with its correct SI unit?

| | quantity | unit |
|---|---|---|
| **A** | current | A |
| **B** | mass | g |
| **C** | temperature | °C |
| **D** | weight | N |

**52 9702/12/O/N/10/Q2**

The frictional force $F$ on a sphere falling through a fluid is given by the formula $F = 6\pi a \eta v$

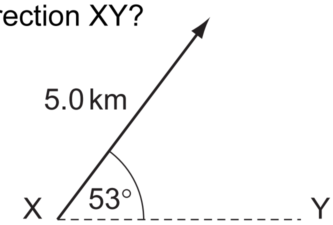
where $a$ is the radius of the sphere, $\eta$ is a constant relating to the fluid and $v$ is the velocity of the sphere.

What are the units of $\eta$?

**A** $\mathrm{kg\,m\,s^{-1}}$ **B** $\mathrm{kg\,m^{-1}\,s^{-1}}$ **C** $\mathrm{kg\,m\,s^{-3}}$ **D** $\mathrm{kg\,m^{3}\,s^{-3}}$

**53 9702/12/O/N/10/Q3**

What is the component of this displacement vector in the direction XY?




**A** 3.0 km **B** 4.0 km

**C** 5.0 km **D** 6.6 km

**54 9702/11/M/J/11/Q1**

Decimal sub-multiples and multiples of units are indicated using a prefix to the unit. For example, the prefix milli (m) represents $10^{-3}$.

Which row gives the sub-multiples or multiples represented by pico (p) and giga (G)?

| | pico (p) | giga (G) |
|---|---|---|
| **A** | $10^{-9}$ | $10^{9}$ |
| **B** | $10^{-9}$ | $10^{12}$ |
| **C** | $10^{-12}$ | $10^{9}$ |
| **D** | $10^{-12}$ | $10^{12}$ |

**55 9702/11/M/J/11/Q2***

Which definition is correct and uses only quantities rather than units?

**A** Density is mass per cubic metre.

**B** Potential difference is energy per unit current.

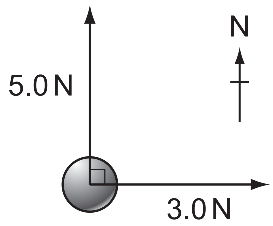
**C** Pressure is force per unit area.

**D** Speed is distance travelled per second.

**56 9702/11/M/J/11/Q3**

A force of 5.0 N pushes a ball due north and another force of 3.0 N pushes it due east.

What is the magnitude of the net force acting on the ball?




**A** 2.8 N **B** 4.0 N

**C** 5.8 N **D** 8.0 N

**57 9702/12/M/J/11/Q1**

Stress has the same SI base units as

**A** $\dfrac{\text{force}}{\text{mass}}$. **B** $\dfrac{\text{force}}{\text{length}}$. **C** $\dfrac{\text{force}}{\text{area}}$. **D** energy.

**58 9702/12/M/J/11/Q2**

To check calculations, the units are put into the following equations together with the numbers. Which equation must be **incorrect**?

**A** force = 300 J / 6 m **C** time = 6 m / 30 m s$^{-1}$

**B** power = 6000 J × 20 s **D** velocity = 4 m s$^{-2}$ × 30 s

**59** **9702/12/M/J/11/Q3**

In making reasonable estimates of physical quantities, which statement is **not** correct?

**A** The frequency of sound can be of the order of GHz.

**B** The wavelength of light can be of the order of 600 nm.

**C** The Young modulus can be of the order of $10^{11}$ Pa.

**D** Beta radiation is associated with one unit of negative charge.

**60** **9702/11/O/N/11/Q1**

Which statement using prefixes of the base unit metre (m) is **not** correct?

**A** 1 pm = $10^{-12}$ m **B** 1 nm = $10^{-9}$ m **C** 1 Mm = $10^{6}$ m **D** 1 Gm = $10^{12}$ m

**61** **9702/01/M/J/05/Q3**

An Olympic athlete of mass 80 kg competes in a 100 m race.

What is the best estimate of his mean kinetic energy during the race?

**A** $4 \times 10^{2}$ J **B** $4 \times 10^{3}$ J **C** $4 \times 10^{4}$ J **D** $4 \times 10^{5}$ J

**62** **9702/11/O/N/11/Q3**

Which group of quantities contains only vectors?

**A** acceleration, displacement, speed

**B** acceleration, work, electric field strength

**C** displacement, force, velocity

**D** power, electric field strength, force

**63** **9702/13/M/J/14/Q1***

Which quantity can be measured in electronvolts (eV)?

**A** electric charge **B** electric potential **C** energy **D** power

**64** **9702/12/O/N/11/Q3**

The following physical quantities can be either positive or negative.

$s$ : displacement of a particle along a straight line
$\theta$ : temperature on the Celsius scale
$q$ : electric charge
$V$ : readings on a digital voltmeter

Which of these quantities are vectors?

**A** $s$, $\theta$, $q$, $V$ **B** $s$, $q$, $V$ only **C** $\theta$, $V$ only **D** $s$ only

**65** **9702/12/M/J/12/Q1**

What is the unit watt in terms of SI base units?

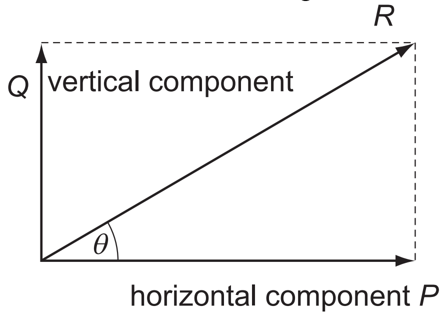
**A** $\mathrm{J\,s^{-1}}$ **B** $\mathrm{m^2\,kg\,s^{-1}}$ **C** $\mathrm{m^2\,kg\,s^{-3}}$ **D** $\mathrm{N\,m\,s^{-1}}$

**66** **9702/12/M/J/12/Q3**

A vector has magnitude $R$ and perpendicular components $P$ and $Q$, as shown in the diagram.

Which row correctly describes the perpendicular components?

| | vertical component | horizontal component |
|---|---|---|
| **A** | $Q$ | $R\sin\theta$ |
| **B** | $R\cos\theta$ | $P$ |
| **C** | $R\cos\theta$ | $R\sin\theta$ |
| **D** | $R\sin\theta$ | $R\cos\theta$ |




**67** **9702/13/M/J/12/Q1**

What is a reasonable estimate of the average kinetic energy of an athlete during a 100 m race that takes 10 s?

**A** 40 J **B** 400 J **C** 4000 J **D** 40 000 J

**68** **9702/13/O/N/15/Q3**

When a force $F$ moves its point of application through a displacement $s$ in the direction of the force, the work $W$ done by the force is given by $W = Fs$.

How many vector quantities and scalar quantities does this equation contain?

**A** one scalar quantity and two vector quantities
**B** one vector quantity and two scalar quantities
**C** three scalar quantities
**D** three vector quantities

**69** **9702/13/M/J/12/Q3**

What is a possible unit for the product $VI$, where $V$ is the potential difference across a resistor and $I$ is the current through the same resistor?

**A** newton per second ($N\,s^{-1}$)
**B** newton second ($N\,s$)
**C** newton metre ($N\,m$)
**D** newton metre per second ($N\,m\,s^{-1}$)

**70** **9702/12/O/N/12/Q1**

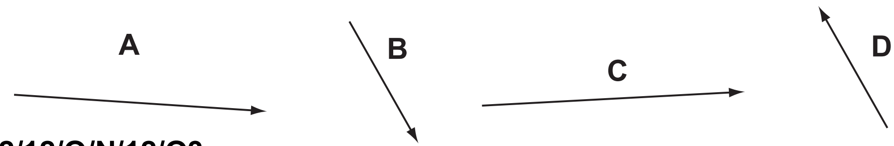
Which quantity has the same base units as momentum?

**A** density × energy **B** density × volume × velocity **C** pressure × area **D** weight ÷ area

**71** **9702/12/O/N/12/Q2**

Vectors P and Q are drawn to scale.


Which diagram represents the vector (P + Q)?




**72** **9702/12/O/N/12/Q3**

What is the approximate kinetic energy of an Olympic athlete when running at maximum speed during a 100 m race?

**A** 400 J **B** 4000 J **C** 40 000 J **D** 400 000 J

**73** **9702/12/O/N/12/Q4**

Physical quantities can be classed as vectors or as scalars.
Which pair of quantities are both vectors?

**A** kinetic energy and elastic force
**B** momentum and time
**C** velocity and electric field strength
**D** weight and temperature

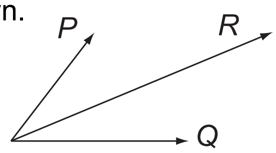
**74** **9702/13/O/N/12/Q1**

The units of all physical quantities can be expressed in terms of SI base units.
Which pair contains quantities with the same base units?

**A** force and momentum
**B** pressure and Young modulus
**C** power and kinetic energy
**D** mass and weight

**75** **9702/13/O/N/12/Q2**

Two physical quantities $P$ and $Q$ are added. The sum of $P$ and $Q$ is $R$, as shown.




Which quantity could be represented by $P$ and by $Q$?

**A** kinetic energy **B** power **C** speed **D** velocity

**76** **9702/13/O/N/12/Q4**

Three of these quantities have the same unit.

Which quantity has a different unit?

**A** $\dfrac{\text{energy}}{\text{distance}}$ **B** force **C** power × time **D** rate of change of momentum

**77** **9702/11/M/J/13/Q1**

Which pair of quantities contains one vector and one scalar quantity?

**A** displacement; force

**B** kinetic energy; power

**C** acceleration; momentum

**D** velocity; distance

**78** **9702/11/M/J/13/Q2**

One property $Q$ of a material is used to describe the behaviour of sound waves in the material. $Q$ is defined as the pressure $P$ of the sound wave divided by the speed $v$ of the wave and the surface area $A$ of the material through which the wave travels:

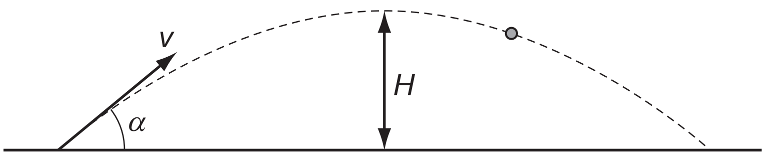
$$Q = \frac{P}{vA}.$$

What are the SI base units of $Q$?

**A** $\text{kg m}^{2}\,\text{s}^{-3}$ **B** $\text{kg m}^{-3}\,\text{s}^{-1}$ **C** $\text{kg m}^{-4}\,\text{s}^{-1}$ **D** $\text{kg m}^{-2}\,\text{s}^{-2}$

**79** **9702/11/M/J/13/Q3**

A cannon fires a cannonball with an initial speed $v$ at an angle $\alpha$ to the horizontal.




Which equation is correct for the maximum height $H$ reached?

**A** $H = \dfrac{v\sin\alpha}{2g}$ **B** $H = \dfrac{g\sin\alpha}{2v}$ **C** $H = \dfrac{(v\sin\alpha)^2}{2g}$ **D** $H = \dfrac{g^2\sin\alpha}{2v}$

**80** **9702/11/M/J/13/Q4**

A wave has a frequency of 5 GHz.

What is the period of the wave?

**A** 20 000 µs **B** 20 ns **C** 2 ns **D** 200 ps

**81** **9702/12/M/J/13/Q1**

Which pair includes a vector quantity and a scalar quantity?

**A** displacement; acceleration

**B** force; kinetic energy

**C** power; speed

**D** work; potential energy

**82** **9702/12/M/J/13/Q2**

The unit of resistivity, expressed in terms of base units, is given by

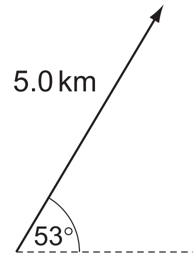
$$\text{kg}\,x^{3}\,y^{-2}\,z^{-3}.$$

Which base units are x, y and z?

| | x | y | z |
|---|---|---|---|
| **A** | ampere | metre | second |
| **B** | metre | ampere | second |
| **C** | metre | second | ampere |
| **D** | second | ampere | metre |

**83 9702/13/M/J/13/Q1**

The diagram shows a displacement vector.




What is the vertical component of this displacement vector?

**A** 3.0 km **B** 4.0 km **C** 5.0 km **D** 6.6 km

**84 9702/13/M/J/13/Q2**

What is the unit of power, expressed in SI base units?

**A** $kg\,m^2\,s^{-3}$ **B** $kg\,m\,s^{-3}$ **C** $kg\,m\,s^{-2}$ **D** $kg\,m^2\,s^{-1}$

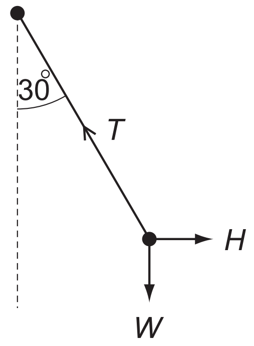
**85 9702/12/O/N/13/Q1**

Which row shows an SI base quantity with its correct unit?

| | SI base quantity | unit |
|---|---|---|
| **A** | charge | coulomb |
| **B** | current | ampere |
| **C** | potential difference | volt |
| **D** | temperature | degree Celsius |

**86 9702/12/O/N/13/Q2**

A pendulum bob is held stationary by a horizontal force $H$. The three forces acting on the bob are shown in the diagram.




The tension in the string of the pendulum is $T$

. The weight of the pendulum bob is $W$.

Which statement is correct?

**A** $H = T\cos 30°$ **C** $W = T\cos 30°$

**B** $T = H\sin 30°$ **D** $W = T\sin 30°$

**87 9702/12/O/N/13/Q3***

The drag coefficient $C_d$ is a number with no units. It is used to compare the drag on different cars at different speeds. It is given by the equation

$$C_d = \frac{2F}{\rho v^n A}$$

where $F$ is the drag force on the car, $\rho$ is the density of the air, $A$ is the cross-sectional area of the car and $v$ is the speed of the car.

What is the value of $n$?

**A** 1 **B** 2 **C** 3 **D** 4

**88 9702/13/O/N/13/Q1**

Which estimate is realistic?

**A** The kinetic energy of a bus travelling on an expressway is 30 000 J.

**B** The power of a domestic light is 300 W.

**C** The temperature of a hot oven is 300 K.

**D** The volume of air in a car tyre is $0.03\,m^3$.

**89 9702/13/O/N/13/Q2***

Which unit is equivalent to the coulomb?

**A** ampere per second **C** watt per ampere

**B** joule per volt **D** watt per volt

**90** **9702/13/O/N/13/Q3***

Two forces of equal magnitude are represented by two coplanar vectors. One is directed eastwards and the other is directed northwards.

What is the direction of a single force that will balance these two forces?

**A** towards the north-east
**B** towards the north-west
**C** towards the south-east
**D** towards the south-west

**91** **9702/13/O/N/13/Q4**

The spring constant $k$ of a coiled wire spring is given by the equation $k = \dfrac{Gr^4}{4nR^3}$

where $r$ is the radius of the wire, $n$ is the number of turns of wire and $R$ is the radius of each of the turns of wire. The quantity $G$ depends on the material from which the wire is made.

What is a suitable unit for $G$?

**A** $\mathrm{N\,m^{-2}}$ **B** $\mathrm{N\,m^{-1}}$ **C** $\mathrm{N\,m}$ **D** $\mathrm{N\,m^{2}}$

**92** **9702/11/M/J/14/Q1**

Which pair of units contains one derived unit and one SI base unit?

**A** ampere coulomb
**B** kilogram kelvin
**C** metre second
**D** newton pascal

**93** **9702/11/M/J/14/Q2**

What is equivalent to 2000 microvolts?

**A** $2\,\mu\mathrm{J\,C^{-1}}$ **B** $2\,\mathrm{mV}$ **C** $2\,\mathrm{pV}$ **D** $2000\,\mathrm{mV}$

**94** **9702/11/M/J/14/Q3**

The speed $v$ of a liquid leaving a tube depends on the change in pressure $\Delta P$ and the density $\rho$ of the liquid. The speed is given by the equation

$$v = k\left(\frac{\Delta P}{\rho}\right)^n$$

where $k$ is a constant that has no units.

What is the value of $n$?

**A** $\frac{1}{2}$ **B** 1 **C** $\frac{3}{2}$ **D** 2

**95** **9702/12/M/J/14/Q1***

The maximum theoretical power $P$ of a wind turbine is given by the equation $P = k\rho Av^n$

where $\rho$ is the density of air, $A$ is the area swept by the turbine blades, $v$ is the speed of the air and $k$ is a constant with no units.

What is the value of $n$?

**A** 1 **B** 2 **C** 3 **D** 4

**96** **9702/12/M/J/14/Q2**

What is the unit of resistance when expressed in SI base units?

**A** $\mathrm{kg\,m^2\,s^{-2}\,A^{-1}}$ **B** $\mathrm{kg\,m^2\,s^{-3}\,A^{-2}}$ **C** $\mathrm{kg\,m\,s^{-2}\,A^{-1}}$ **D** $\mathrm{kg\,m\,s^{-3}\,A^{-1}}$

**97** **9702/12/O/N/11/Q1**

Which quantity can be measured in electronvolts (eV)?

**A** electric charge **B** electric potential **C** energy **D** power

**98** **9702/13/M/J/14/Q2**

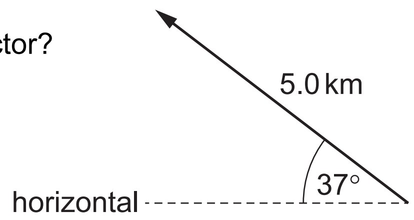
The unit of specific heat capacity is $\mathrm{J\,kg^{-1}\,K^{-1}}$.

What is its equivalent in terms of SI base units?

**A** $\mathrm{kg^{-1}\,m^2\,K^{-1}}$ **B** $\mathrm{m\,s^{-1}\,K^{-1}}$ **C** $\mathrm{m\,s^{-2}\,K^{-1}}$ **D** $\mathrm{m^2\,s^{-2}\,K^{-1}}$

**99** **9702/13/M/J/14/Q3***

What is the vertical component of this displacement vector?




**A** 3.0 km **B** 3.8 km

**C** 4.0 km **D** 5.0 km

**100** **9702/12/O/N/14/Q1**

A 0.10 kg mass is taken to Mars and then weighed on a spring balance and on a lever balance. The acceleration due to gravity on Mars is 38% of its value on Earth.

What are the readings on the two balances on Mars? (Assume that on Earth $g = 10\,\mathrm{m\,s^{-2}}$.)

| | spring balance / N | lever balance / kg |
|---|---|---|
| **A** | 0.38 | 0.038 |
| **B** | 0.38 | 0.10 |
| **C** | 1.0 | 0.038 |
| **D** | 1.0 | 0.10 |

**101** **9702/11/O/N/14/Q2**

What is equivalent to the unit of electric field strength?

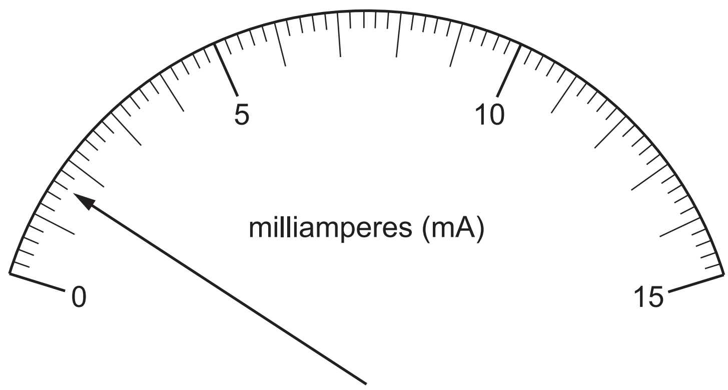
**A** $\mathrm{J\,C\,m^{-1}}$ **B** $\mathrm{N\,s\,A^{-1}}$ **C** $\mathrm{kg\,m\,s^{-3}\,A^{-1}}$ **D** $\mathrm{kg\,m^{3}\,s^{-3}\,A^{-1}}$

**102** **9702/11/O/N/14/Q3**

The diagram shows the reading on an analogue ammeter.

Which digital ammeter reading is the same as the reading on the analogue ammeter?

| | display units | display reading |
|---|---|---|
| **A** | μA | 1600 |
| **B** | μA | 160 |
| **C** | mA | 16.0 |
| **D** | A | 1.60 |




**103** **9702/13/O/N/14/Q1**

When the brakes are applied on a vehicle moving at speed $v$, the distance $d$ moved by the vehicle in coming to rest is given by the expression

$$d = kv^2$$

where $k$ is a constant.

What is the unit of $k$ expressed in SI base units?

**A** $\mathrm{m^{-1}\,s^{2}}$ **B** $\mathrm{m\,s^{-2}}$ **C** $\mathrm{m^{2}\,s^{-2}}$ **D** $\mathrm{m^{-1}\,s}$

**104** **9702/13/O/N/14/Q2**

Which list contains one vector quantity and two scalar quantities?

**A** displacement, weight, velocity

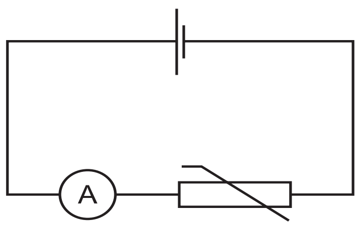
**B** force, acceleration, time

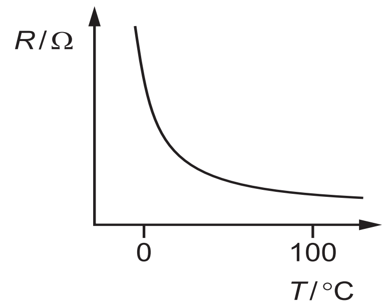
**C** momentum, mass, speed

**D** work, density, energy

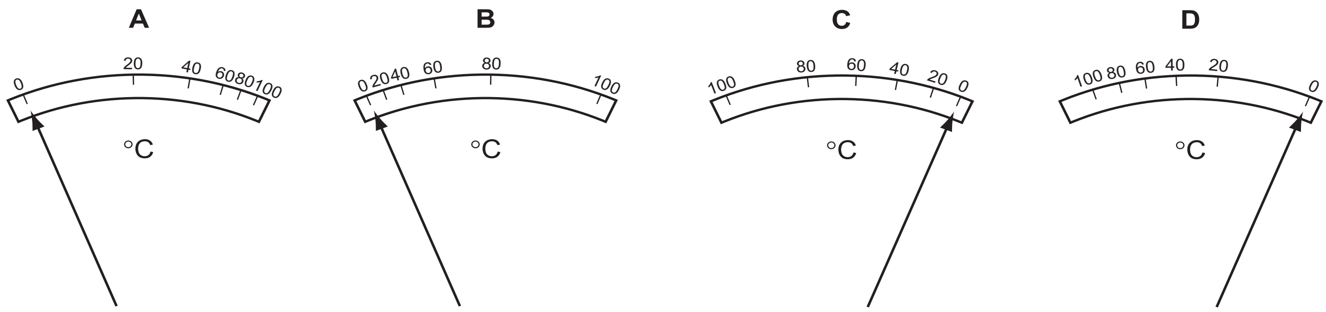
**105** **9702/13/O/N/14/Q3**

In the circuit shown, an analogue ammeter is to be recalibrated as a thermometer. The graph shows how the resistance $R$ of the thermistor changes with temperature $T$.




Which diagram could represent the temperature scale on the ammeter?




**106** **9702/11/M/J/15/Q1**

Which is an SI base unit?

**A** current **B** gram **C** kelvin **D** volt

**107** **9702/11/M/J/15/Q2**

Which pair contains one vector and one scalar quantity?

| | | | |
|---|---|---|---|
| **A** | displacement | acceleration |
| **B** | force | kinetic energy |
| **C** | momentum | velocity |
| **D** | power | speed |

**108** **9702/11/M/J/15/Q3**

When a constant braking force is applied to a vehicle moving at speed $v$, the distance $d$ moved by the vehicle in coming to rest is given by the expression

$$d = kv^2$$

where $k$ is a constant.

When $d$ is measured in metres and $v$ is measured in metres per second, the constant has a value of $k_1$.

What is the value of the constant when the distance is measured in metres, and the speed is measured in kilometres per hour?

**A** $0.0772\,k_1$ **B** $0.278\,k_1$ **C** $3.60\,k_1$ **D** $13.0\,k_1$

**109 9702/12/M/J/15/Q1**

Which definition is correct and uses only quantities rather than units?

**A** Density is mass per cubic metre.

**B** Potential difference is energy per unit current.

**C** Pressure is force per unit area.

**D** Speed is distance travelled per second.

**110 9702/12/M/J/15/Q2**

The average kinetic energy $E$ of a gas molecule is given by the equation $E = \frac{3}{2}kT$

where $T$ is the absolute (kelvin) temperature.

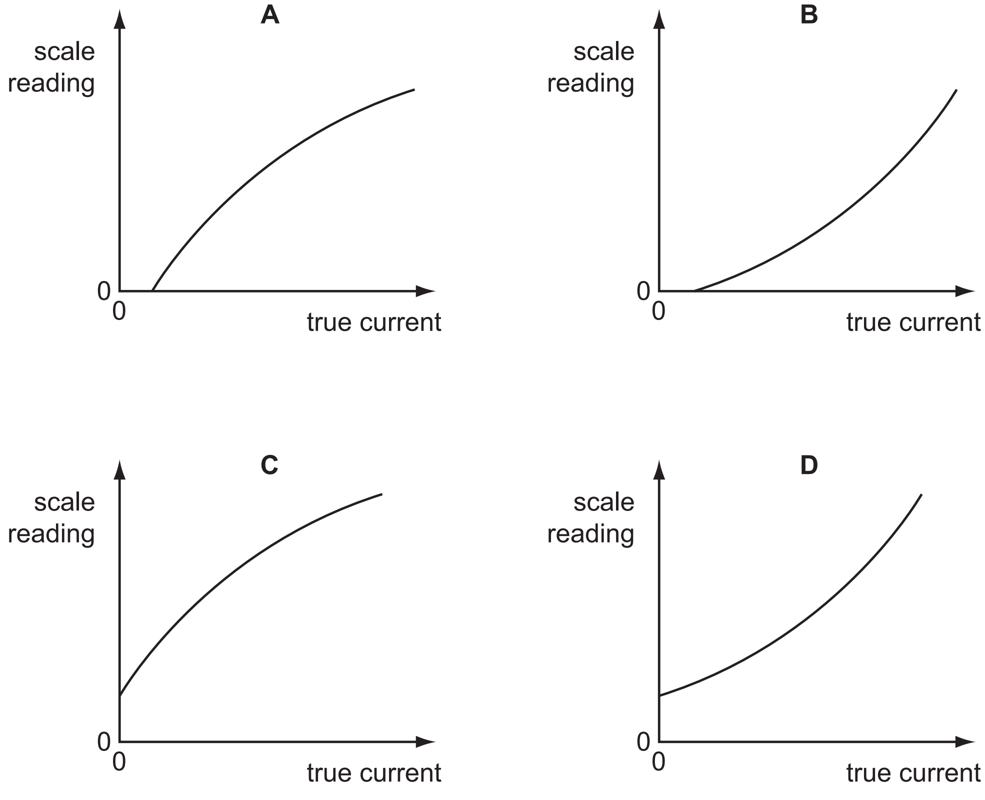
What are the SI base units of $k$?

**A** $\mathrm{kg^{-1}\,m^{-1}\,s^{2}\,K}$ **B** $\mathrm{kg^{-1}\,m^{-2}\,s^{2}\,K}$ **C** $\mathrm{kg\,m\,s^{-2}\,K^{-1}}$ **D** $\mathrm{kg\,m^{2}\,s^{-2}\,K^{-1}}$

**111 9702/12/M/J/15/Q3**

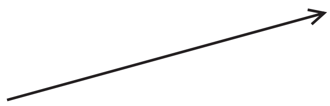
An analogue ammeter has a pointer which moves over a scale. Following prolonged use, the pointer does not return fully to zero when the current is turned off and the meter has become less sensitive at higher currents than it is at lower currents.

Which diagram best represents the calibration graph needed to obtain an accurate current reading?

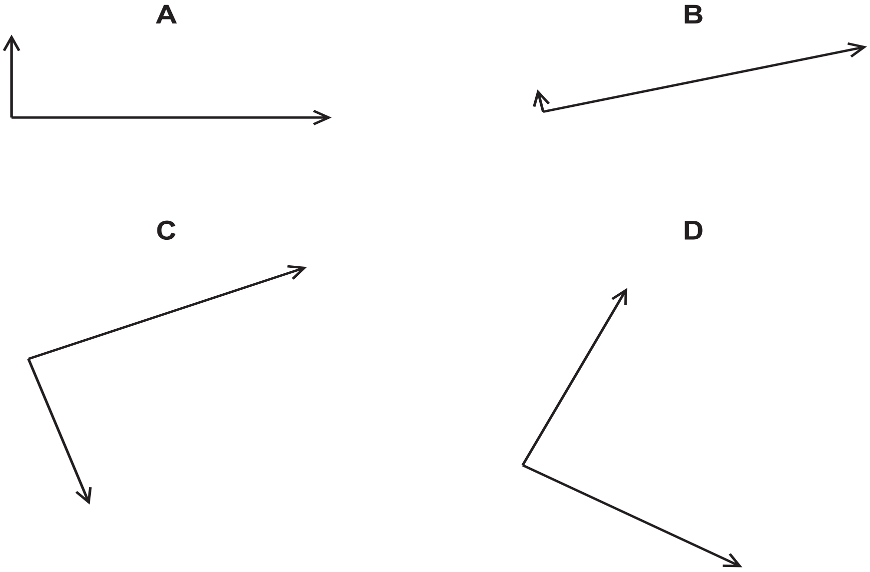


**112** **9702/12/M/J/15/Q4**

The arrow represents the vector R.



Which diagram does **not** represent R as two perpendicular components?




**113** **9702/13/M/J/15/Q1**

Which statement includes a correct unit?

**A** energy = 7.8 N s

**B** force = 3.8 N s

**C** momentum = 6.2 N s

**D** torque = 4.7 N s

**114** **9702/13/M/J/15/Q2**

What is the joule (J) in SI base units?

**A** $\mathrm{kg\,m\,s^{-1}}$ **B** $\mathrm{kg\,m^2\,s^{-1}}$ **C** $\mathrm{kg\,m\,s^{-2}}$ **D** $\mathrm{kg\,m^2\,s^{-2}}$

**115** **9702/13/M/J/15/Q3**

The speed of an aeroplane in still air is $200\,\mathrm{km\,h^{-1}}$. The wind blows from the west at a speed of $85.0\,\mathrm{km\,h^{-1}}$.

In which direction must the pilot steer the aeroplane in order to fly due north?

**A** 23.0° east of north

**B** 23.0° west of north

**C** 25.2° east of north

**D** 25.2° west of north

**116** **9702/11/M/J/17/Q2**

A particle travels in a straight line with speed $v$.

The particle slows down and changes direction. The new speed of the particle is $\frac{v}{2}$.

The new velocity has a component of $\frac{v}{4}$ in the same direction as the initial path of the particle.

Through which angle has the particle turned?

**A** 27° **B** 30° **C** 45° **D** 60°

**117** **9702/12/M/J/17/Q3**

What correctly expresses the volt in terms of SI base units?

**A** $\mathrm{A\,\Omega}$ **B** $\mathrm{W\,A^{-1}}$ **C** $\mathrm{kg\,m^2\,s^{-1}\,A^{-1}}$ **D** $\mathrm{kg\,m^2\,s^{-3}\,A^{-1}}$

**118** **9702/11/O/N/17/Q1**

Which SI unit, expressed in base units, is **not** correct?

**A** unit of force, $\mathrm{kg\,m\,s^{-2}}$

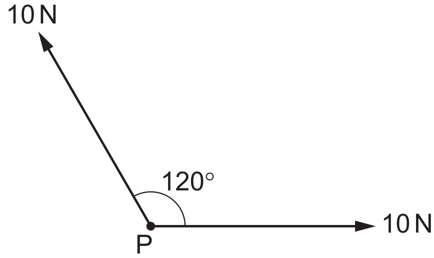
**B** unit of momentum, $\mathrm{kg\,m\,s^{-1}}$

**C** unit of pressure, $\mathrm{kg\,m^{-2}\,s^{-2}}$

**D** unit of work, $\mathrm{kg\,m^2\,s^{-2}}$

**119** **9702/11/O/N/17/Q2**

Two forces, each of 10 N, act at a point P as shown. The angle between the directions of the forces is 120°.




What is the magnitude of the resultant force?

**A** 5 N

**B** 10 N

**C** 17 N

**D** 20 N

**120** **9702/11/O/N/17/Q6**

In still air, a bird can fly at a speed of $10\,\mathrm{m\,s^{-1}}$. The wind is blowing from the east at $8.0\,\mathrm{m\,s^{-1}}$.

In which direction must the bird fly in order to travel to a destination that is due north of the bird's current location?

**A** 37° east of north

**B** 37° west of north

**C** 53° east of north

**D** 53° west of north

**121** **9702/12/O/N/17/Q1**

Which pair of units are **not** the same when expressed in SI base units?

**A** $\mathrm{m\,s^{-2}}$ and $\mathrm{N\,kg^{-1}}$

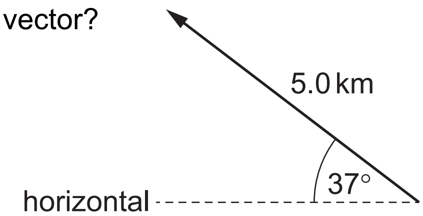
**B** $\mathrm{N\,s}$ and $\mathrm{kg\,m\,s^{-1}}$

**C** Pa and $\mathrm{N\,m^{-2}}$

**D** $\mathrm{V\,m^{-2}}$ and $\mathrm{N\,C^{-1}}$

**122** **9702/12/O/N/17/Q2**

What is the vertical component of this displacement vector?




**A** 3.0 km **B** 3.8 km

**C** 4.0 km **D** 5.0 km

**123** **9702/12/O/N/17/Q3**

The units of specific heat capacity are $\mathrm{J\,kg^{-1}\,K^{-1}}$.

What are the SI base units of specific heat capacity?

**A** $\mathrm{m\,s^{-2}\,K^{-1}}$ **B** $\mathrm{m\,s^{-1}\,K^{-1}}$ **C** $\mathrm{m^2\,s^{-2}\,K^{-1}}$ **D** $\mathrm{m^2\,s^{-1}\,K^{-1}}$

**124 9702/12/O/N/17/Q4**

A quantity $y$ is to be determined from the equation shown. $y = \frac{px}{q^2}$

The percentage uncertainties in $p$, $x$ and $q$ are shown.

What is the percentage uncertainty in $y$?

| | percentage uncertainty |
|---|---|
| $p$ | 6% |
| $x$ | 2% |
| $q$ | 4% |

**A** 0.5% **B** 0.75%

**C** 12% **D** 16%

**125 9702/13/O/N/17/Q1**

How many cubic nanometres, $nm^3$, are in a cubic micrometre, $\mu m^3$?

**A** $10^3$ **B** $10^6$ **C** $10^9$ **D** $10^{12}$

**126 9702/13/O/N/17/Q2**

The maximum theoretical power $P$ of a wind turbine is given by the equation $P = k\rho Av^n$

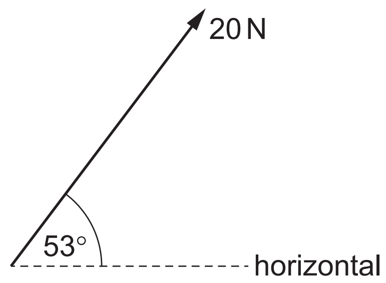
where $\rho$ is the density of air, $A$ is the area swept by the turbine blades, $v$ is the speed of the air and $k$ is a constant with no units.

What is the value of $n$?

**A** 1 **B** 2 **C** 3 **D** 4

**127 9702/13/O/N/17/Q3**

What is the horizontal component of the force shown?




**A** 12 N

**B** 16 N

**C** 20 N

**D** 27 N

**128 9702/11/M/J/18/Q1**

What is a unit for stress?

**A** $kg\,m^{-1}\,s^{-2}$ **B** $kg\,m^{-2}\,s^{-2}$ **C** $N\,m^{-1}$ **D** $N\,m$

**129 9702/11/M/J/18/Q2**

Physical quantities can be classed as vectors or as scalars.

Which pair of quantities consists of two vectors?

**A** kinetic energy and force

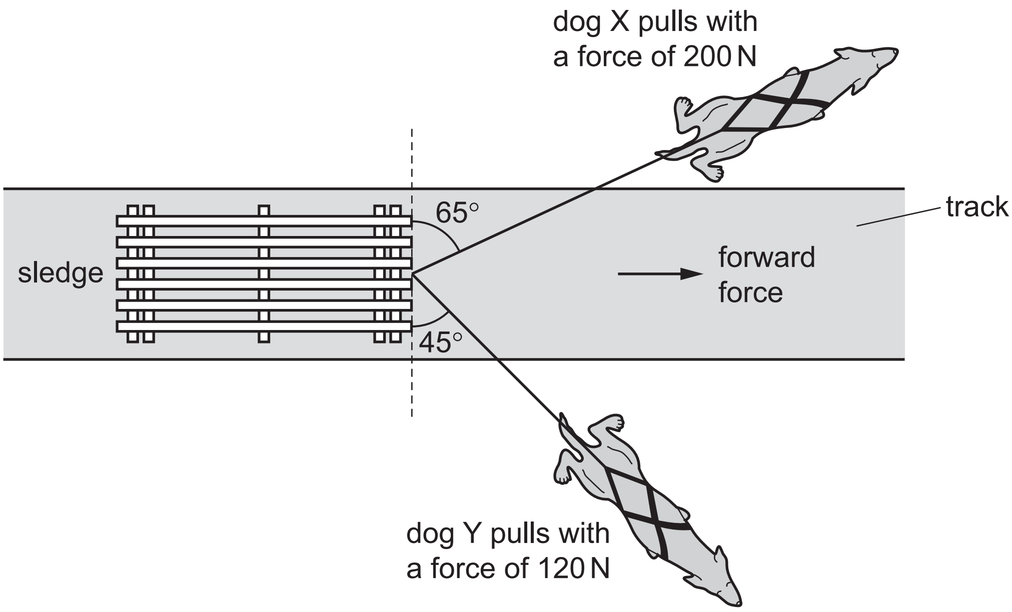
**B** momentum and time

**C** velocity and electric field strength

**D** weight and temperature

**130** **9702/11/M/J/18/Q3**

Two dogs pull a sledge along an icy track, as shown.




Dog X pulls with a force of 200 N at an angle of 65° to the front edge of the sledge. Dog Y pulls with a force of 120 N at an angle of 45° to the front edge of the sledge.

What is the resultant forward force on the sledge exerted by the two dogs?

**A** 80 N **B** 170 N **C** 270 N **D** 320 N

**131** **9702/12/M/J/18/Q1**

A sheet of gold leaf has a thickness of 0.125 μm. A gold atom has a radius of 174 pm.

Approximately how many layers of atoms are there in the sheet?

**A** 4 **B** 7 **C** 400 **D** 700

**132** **9702/12/M/J/18/Q2**

The drag coefficient $C_d$ is a number with no units. It is used to compare the drag on different cars at different speeds. $C_d$ is given by the equation

$$C_d = \frac{2F}{v^n \rho A}$$

where $F$ is the drag force on the car, $\rho$ is the density of the air, $A$ is the cross-sectional area of the car and $v$ is the speed of the car.

What is the value of $n$?

**A** 1 **B** 2 **C** 3 **D** 4

**133 9702/13/M/J/18/Q1**

What is the best way of describing a physical quantity?

**A** a quantity with a magnitude and a direction but no unit
**B** a quantity with a magnitude and a unit
**C** a quantity with a magnitude but no direction
**D** a quantity with a unit but no magnitude

**134 9702/13/M/J/18/Q2**

Which pair includes a vector quantity and a scalar quantity?

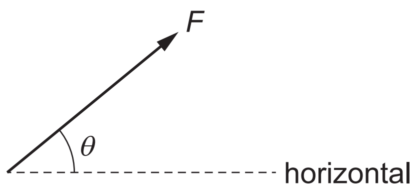
**A** displacement and acceleration
**B** force and kinetic energy
**C** power and speed
**D** work and potential energy

**135 9702/13/M/J/18/Q3**

A force $F$ acts at an angle $\theta$ to the horizontal.

What are the horizontal and the vertical components of the force?

| | horizontal component | vertical component |
|---|---|---|
| **A** | $F\cos\theta$ | $F\cos(90° - \theta)$ |
| **B** | $F\cos\theta$ | $F\sin(90° - \theta)$ |
| **C** | $F\sin\theta$ | $F\cos\theta$ |
| **D** | $F\sin\theta$ | $F\cos(90° - \theta)$ |




**136 9702/12/F/M/18/Q1**

Which unit is equivalent to the coulomb?

**A** ampere per second **B** joule per volt **C** watt per ampere **D** watt per volt

**137 9702/12/F/M/18/Q2**

Which row shows a quantity and an **incorrect** unit?

| | quantity | unit |
|---|---|---|
| **A** | efficiency | no unit |
| **B** | moment of force | $\mathrm{N\,m^{-1}}$ |
| **C** | momentum | $\mathrm{N\,s}$ |
| **D** | work done | J |

**138 9702/12/F/M/18/Q3**

Two forces of equal magnitude are represented by two coplanar vectors. One is directed towards the east and the other is directed towards the north.

What is the direction of a single force that will balance these two forces?

**A** towards the north-east
**B** towards the north-west
**C** towards the south-east
**D** towards the south-west

**139** **9702/11/O/N/18/Q1**

The radius of the Earth is approximately $6.4 \times 10^6$ m, and the radius of the Moon is approximately $1.7 \times 10^6$ m. A student wishes to build a scale model of the Solar System in the classroom, using a football of radius 0.12 m to represent the Earth.

Which object would best represent the Moon?

**A** basketball

**B** cherry

**C** golf ball

**D** tennis ball

**140** **9702/11/O/N/18/Q2**

When a beam of light is incident on a surface, it delivers energy to the surface. The intensity of the beam is defined as the energy delivered per unit area per unit time.

What is the unit of intensity, expressed in SI base units?

**A** $kg\,m^{-2}\,s^{-1}$ **B** $kg\,m^{2}\,s^{-3}$ **C** $kg\,s^{-2}$ **D** $kg\,s^{-3}$

**141** **9702/11/O/N/18/Q3**

A ship is travelling with a velocity of $8.0\,km\,h^{-1}$ in a direction 30° east of north.

What are the components of the ship's velocity in the east and north directions?

| | component of velocity in east direction / $km\,h^{-1}$ | component of velocity in north direction / $km\,h^{-1}$ |
|---|---|---|
| **A** | 4.0 | 4.0 |
| **B** | 4.0 | 6.9 |
| **C** | 4.6 | 6.9 |
| **D** | 6.9 | 4.0 |

**142** **9702/12/O/N/18/Q2**

What is the unit of resistance when expressed in SI base units?

**A** $kg\,m^{2}\,s^{-2}\,A^{-1}$

**B** $kg\,m^{2}\,s^{-3}\,A^{-2}$

**C** $kg\,m\,s^{-2}\,A^{-1}$

**D** $kg\,m\,s^{-3}\,A^{-1}$

**143** **9702/12/O/N/18/Q3**

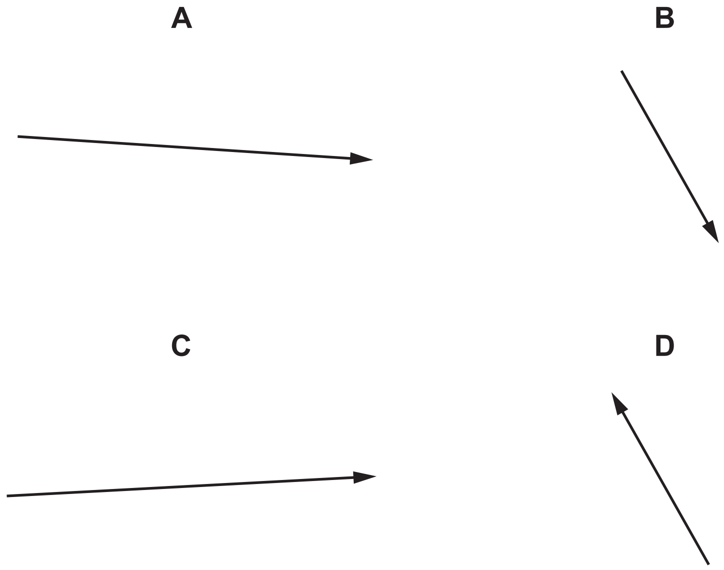
Which list contains both scalar and vector quantities?

**A** acceleration, momentum, velocity, weight
**B** area, current, force, work
**C** distance, kinetic energy, power, pressure
**D** mass, temperature, time, speed

**144** **9702/12/O/N/18/Q4**

Vectors P and Q are drawn to scale.


Which diagram represents the vector (P + Q)?




**145** **9702/13/O/N/18/Q2**

Three of these quantities have the same unit.

Which quantity has a different unit?

**A** $\dfrac{\text{energy}}{\text{distance}}$

**B** force

**C** power $\times$ time

**D** rate of change of momentum

**146** **9702/13/O/N/18/Q3**

Which group of quantities contains only vectors?

**A** acceleration, displacement, speed

**B** acceleration, work, electric field strength

**C** displacement, force, velocity

**D** power, electric field strength, force

**147** **9702/11/M/J/19/Q1**

Which unit can be expressed in base units as $kg\,m^2\,s^{-2}$?

**A** joule

**B** newton

**C** pascal

**D** watt

**148** **9702/11/M/J/19/Q2**

The luminosity $L$ of a star is given by

$$L = 4\pi r^2 \sigma T^4$$

where

$r$ is the radius of the star,

$T$ is the temperature of the star and

$\sigma$ is a constant with units $W\,m^{-2}\,K^{-4}$.

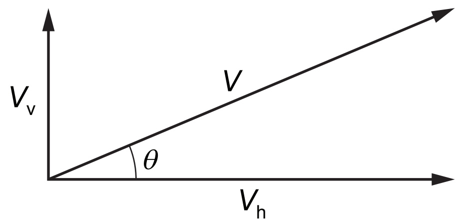
What are the SI base units of $L$?

**A** $kg\,m^2\,s^{-1}$ **B** $kg\,m^2\,s^{-2}$ **C** $kg\,m^2\,s^{-3}$ **D** $kg\,m^2\,s^{-4}$

**149** **9702/11/M/J/19/Q3**

A particle has velocity $V$ at an angle $\theta$ to the horizontal.

The components of the particle's velocity are $V_v$ upwards in the vertical direction and $V_h$ to the right in the horizontal direction, as shown.




What are expressions for the magnitude of $V$ and for the angle $\theta$?

| | magnitude of $V$ | $\theta$ |
|---|---|---|
| **A** | $\sqrt{(V_v^2 + V_h^2)}$ | $\tan^{-1}\left(\frac{V_h}{V_v}\right)$ |
| **B** | $\sqrt{(V_v^2 + V_h^2)}$ | $\tan^{-1}\left(\frac{V_v}{V_h}\right)$ |
| **C** | $\sqrt{(V_v^2 - V_h^2)}$ | $\tan^{-1}\left(\frac{V_h}{V_v}\right)$ |
| **D** | $\sqrt{(V_v^2 - V_h^2)}$ | $\tan^{-1}\left(\frac{V_v}{V_h}\right)$ |

**150 9702/12/M/J/19/Q1**

What is equivalent to 2000 microvolts?

**A** $2\,\mu J\,C^{-1}$ **B** $2\,mV$ **C** $2\,pV$ **D** $2000\,mV$

**151 9702/12/M/J/19/Q2**

What is the number of SI base units required to express electric field strength and power?

| | electric field strength | power |
|---|---|---|
| **A** | 3 | 3 |
| **B** | 3 | 2 |
| **C** | 4 | 2 |
| **D** | 4 | 3 |

**152 9702/12/M/J/19/Q3**

The Planck constant $h$ has SI units J s.

Which equation could be used to calculate the Planck constant?

**A** $h = \dfrac{DE}{v}$ where $D$ is distance, $E$ is energy and $v$ is velocity

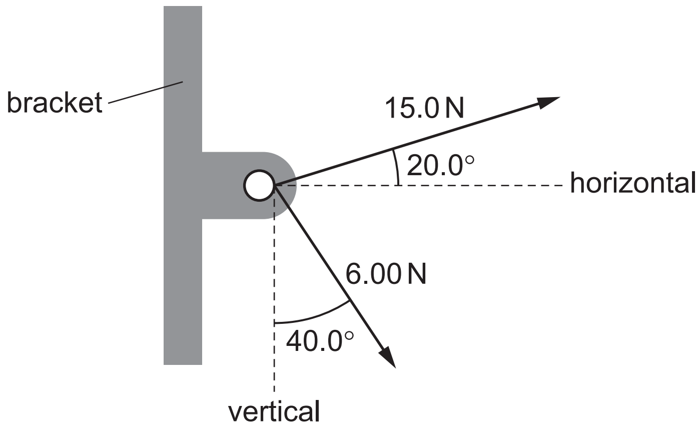
**B** $h = \dfrac{v}{D}$ where $v$ is velocity and $D$ is distance

**C** $h = \dfrac{1}{4\pi E}$ where $E$ is electric field strength

**D** $h = \dfrac{Fr^2}{m}$ where $F$ is force, $r$ is radius and $m$ is mass

**153** **9702/12/M/J/19/Q4**

Two cables are attached to a bracket and exert forces as shown.




What are the magnitudes of the horizontal and vertical components of the resultant of the two forces?

| | horizontal component / N | vertical component / N |
|---|---|---|
| **A** | 9.73 | 0.534 |
| **B** | 9.73 | 10.2 |
| **C** | 18.0 | 0.534 |
| **D** | 18.0 | 10.2 |

**154** **9702/13/M/J/19/Q1**

Which is an SI base unit?

**A** current

**B** gram

**C** kelvin

**D** volt

**155** **9702/13/M/J/19/Q2**

Osmium, a naturally occurring element, has a density of 23 000 kg m$^{-3}$.

What is also a value of the density of osmium?

**A** $2.3 \times 10^{4}\,\mu g\,cm^{-3}$

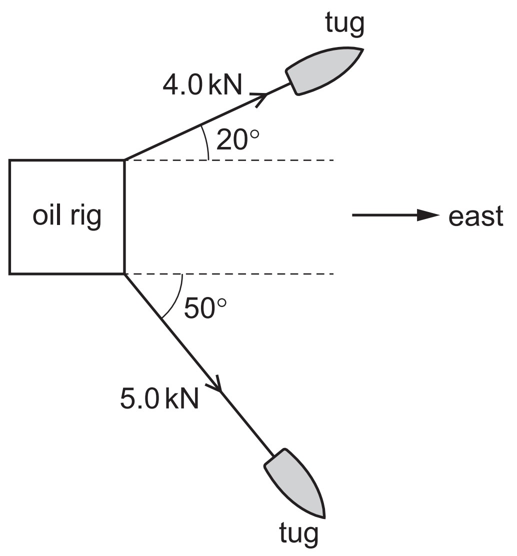
**B** $2.3 \times 10^{4}\,g\,cm^{-3}$

**C** $2.3\,kg\,cm^{-3}$

**D** $2.3 \times 10^{-2}\,kg\,cm^{-3}$

**156 9702/13/M/J/19/Q3**

Two tugs are towing an oil rig as shown.



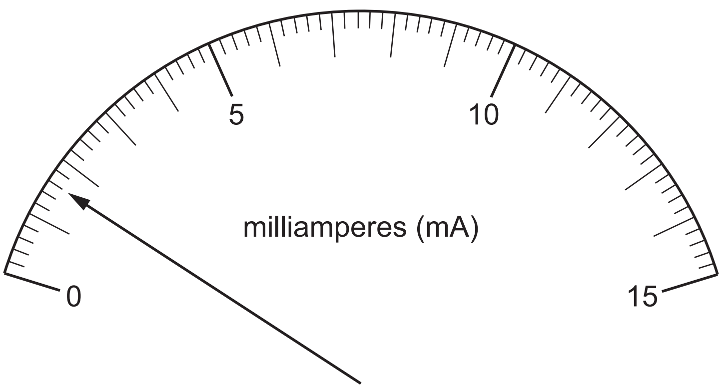

The tensions in the towing cables are 4.0 kN and 5.0 kN.

What is the total force acting on the rig due to the cables, in the direction to the east?

**A** 3.1 kN **B** 5.2 kN **C** 7.0 kN **D** 7.3 kN

**157 9702/13/M/J/19/Q5**

The diagram shows the reading on an analogue ammeter.

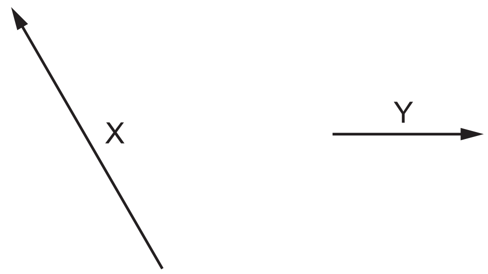



Which digital ammeter reading is the same as the reading on the analogue ammeter?

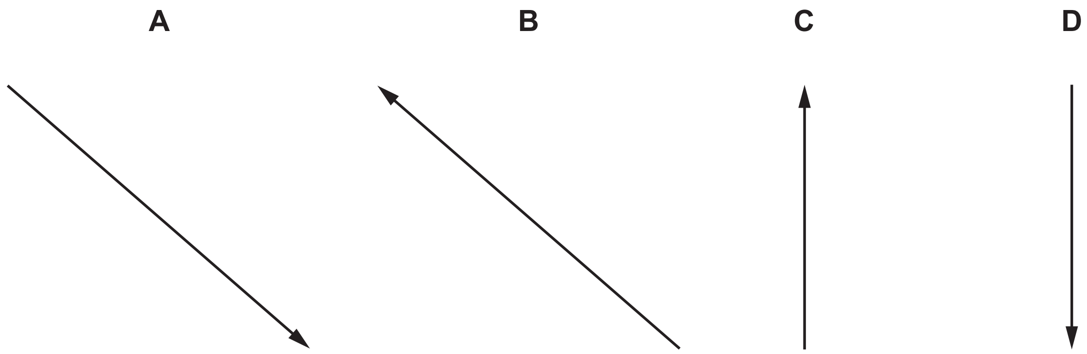
| | display units | display reading |
|---|---|---|
| **A** | μA | 1600 |
| **B** | μA | 160 |
| **C** | mA | 16.0 |
| **D** | A | 1.60 |

**158** **9702/11/O/N/19/Q3**

The diagram shows two vectors X and Y, drawn to scale.




If X = Y – Z, which diagram best represents the vector Z?




**159** **9702/12/O/N/19/Q2**

Which expression gives an SI base quantity?

**A** charge per unit time

**B** force per unit area

**C** mass per unit volume

**D** work done per unit distance

**160** **9702/12/O/N/19/Q3**

Which list contains only scalar quantities?

**A** area, length, displacement

**B** kinetic energy, speed, power

**C** potential energy, momentum, time

**D** velocity, distance, temperature

**161** **9702/13/O/N/19/Q1**

Which quantity with its unit is correct?

**A** acceleration of a bicycle = $1.4\,\mathrm{m\,s^{-1}}$

**B** electric current in a lamp = $0.25\,\mathrm{A\,s^{-1}}$

**C** electric potential difference across a battery = $8.0\,\mathrm{J\,C^{-1}}$

**D** kinetic energy of a car = $4500\,\mathrm{N\,m^{-1}}$

**162** **9702/13/O/N/19/Q2**

Which two units are **not** equivalent to each other?

**A** $\mathrm{N\,m}$ and $\mathrm{kg\,m^2\,s^{-2}}$

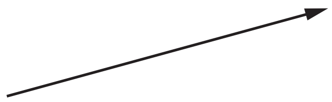
**B** $\mathrm{N\,s}$ and $\mathrm{kg\,m\,s^{-1}}$

**C** $\mathrm{J\,s^{-1}}$ and $\mathrm{kg\,m^2\,s^{-3}}$

**D** $\mathrm{Pa}$ and $\mathrm{kg\,m\,s^{-2}}$

**163** **9702/13/O/N/19/Q3**

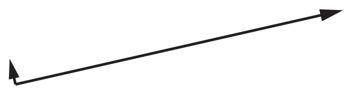
The arrow represents a vector R.



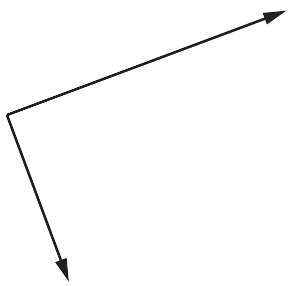
Which diagram does **not** represent R as two perpendicular components?

**A**

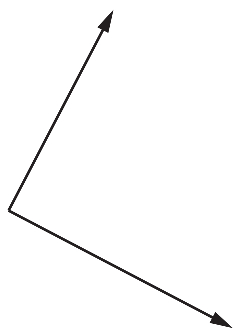
**B**



**C**



**D**



## 164. 9702/12/F/M/20/Q1

The table shows some measurable quantities.

Which row gives the correct order of magnitude of the measurable quantity in the stated unit?

| | measurable quantity | order of magnitude | unit |
|---|---|---|---|
| **A** | mass of a coin | $10^{-4}$ | kg |
| **B** | thickness of a sheet of paper | $10^{-2}$ | m |
| **C** | weight of an apple | $10^{0}$ | N |
| **D** | temperature of a person's body | $10^{1}$ | K |

## 165. 9702/12/F/M/20/Q2

A byte (b) comprises 8 bits.

How many bits are there in 1 terabyte (1Tb)?

**A** $1 \times 10^{9}$ **B** $8 \times 10^{9}$ **C** $1 \times 10^{12}$ **D** $8 \times 10^{12}$

## 166. 9702/12/F/M/20/Q3

Which pair of quantities contains both a scalar **and** a vector?

**A** acceleration and momentum

**B** charge and resistance

**C** kinetic energy and mass

**D** temperature and velocity

## 167. 9702/11/M/J/20/Q1

What is a reasonable estimate of the kinetic energy of a car travelling at a speed of $30\,\mathrm{m\,s^{-1}}$?

**A** $10^{2}$ J **B** $10^{4}$ J **C** $10^{6}$ J **D** $10^{8}$ J

## 168. 9702/11/M/J/20/Q2

The frequency $f$ of vibration of a mass $m$ supported by a spring with spring constant $k$ is given by the equation

$$f = Cm^{p}k^{q}$$

where $C$ is a constant with no units.

What are the values of $p$ and $q$?

| | $p$ | $q$ |
|---|---|---|
| **A** | $-\frac{1}{2}$ | $-\frac{1}{2}$ |
| **B** | $-\frac{1}{2}$ | $\frac{1}{2}$ |
| **C** | $\frac{1}{2}$ | $-\frac{1}{2}$ |
| **D** | $\frac{1}{2}$ | $\frac{1}{2}$ |

**169. 9702/11/M/J/20/Q3**

The power produced by a force moving an object is given by the equation shown.

$$\text{power} = \frac{\text{work}}{\text{time}} = \frac{\text{force} \times \text{displacement}}{\text{time}}$$

Which quantities are scalars and which are vectors?

| | scalars | vectors |
|---|---|---|
| **A** | displacement, time | force, power |
| **B** | power, work | displacement, force |
| **C** | power, force | displacement, work |
| **D** | work, time | power, displacement |

**170. 9702/12/M/J/20/Q1**

What is a reasonable estimate of the mass of a raindrop?

**A** $10^{1}$ kg **B** $10^{-1}$ kg **C** $10^{-3}$ kg **D** $10^{-5}$ kg

**171. 9702/12/M/J/20/Q2**

Which quantity is a scalar?

**A** acceleration

**B** force

**C** kinetic energy

**D** momentum

**172. 9702/13/M/J/20/Q1**

A man is running a race in a straight line.

What is an approximate value of his kinetic energy?

**A** 10 J **B** 100 J **C** 1000 J **D** 10 000 J

**173. 9702/13/M/J/20/Q2**

A sample of gas has a mass of 4.8 μg and occupies a volume of 1.2 $dm^3$.

What is the density of the sample of gas?

**A** $4.0 \times 10^{-3}\,kg\,m^{-3}$

**B** $4.0 \times 10^{-5}\,kg\,m^{-3}$

**C** $4.0 \times 10^{-6}\,kg\,m^{-3}$

**D** $4.0 \times 10^{-8}\,kg\,m^{-3}$

**174. 9702/13/M/J/20/Q3**

Which characteristics are possessed by a vector quantity but **not** by a scalar quantity?

**A** direction only

**B** magnitude and direction

**C** magnitude and unit

**D** unit only

**175. 9702/13/M/J/20/O4**

A circuit is set up in order to determine the resistance of a 12 V, 1.2 W lamp when operating normally. An analogue ammeter and an analogue voltmeter are used.

Which ranges for the meters would be most suitable?

| | ammeter range / A | voltmeter range / V |
|---|---|---|
| **A** | 0–0.5 | 0–20 |
| **B** | 0–0.5 | 0–100 |
| **C** | 0–10 | 0–20 |
| **D** | 0–10 | 0–100 |

**176. 9702/11/O/N/20/Q1**

Which quantity is a physical quantity?

**A** atomic number

**B** efficiency

**C** number density of charge carriers

**D** strain

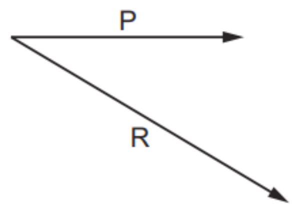
**177. 9702/11/O/N/20/Q2**

Which time interval is the shortest?

**A** 0.05 ms **B** 50 ns **C** 500 000 ps **D** 0.5 $\mu$s

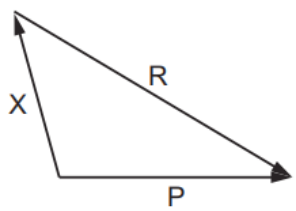
**178. 9702/11/O/N/20/Q3**

P and R are coplanar vectors.

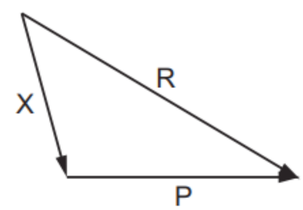



If X = P – R, which diagram best represents vector X?

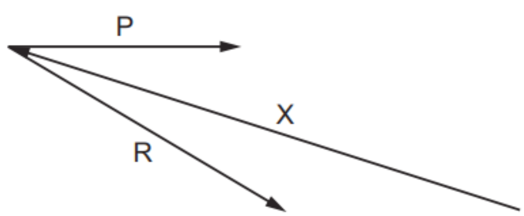
**A**



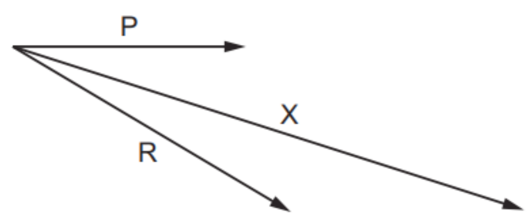

**B**




**C**




**D**




**179. 9702/12/O/N/20/Q2**

The speed $v$ of waves on a stretched wire is given by the equation

$$v = T^p \mu^q$$

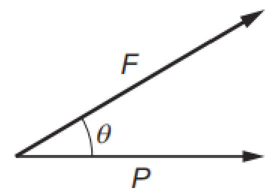
where $T$ is the tension in the wire and $\mu$ is the mass per unit length of the wire.

What are the values of $p$ and $q$?

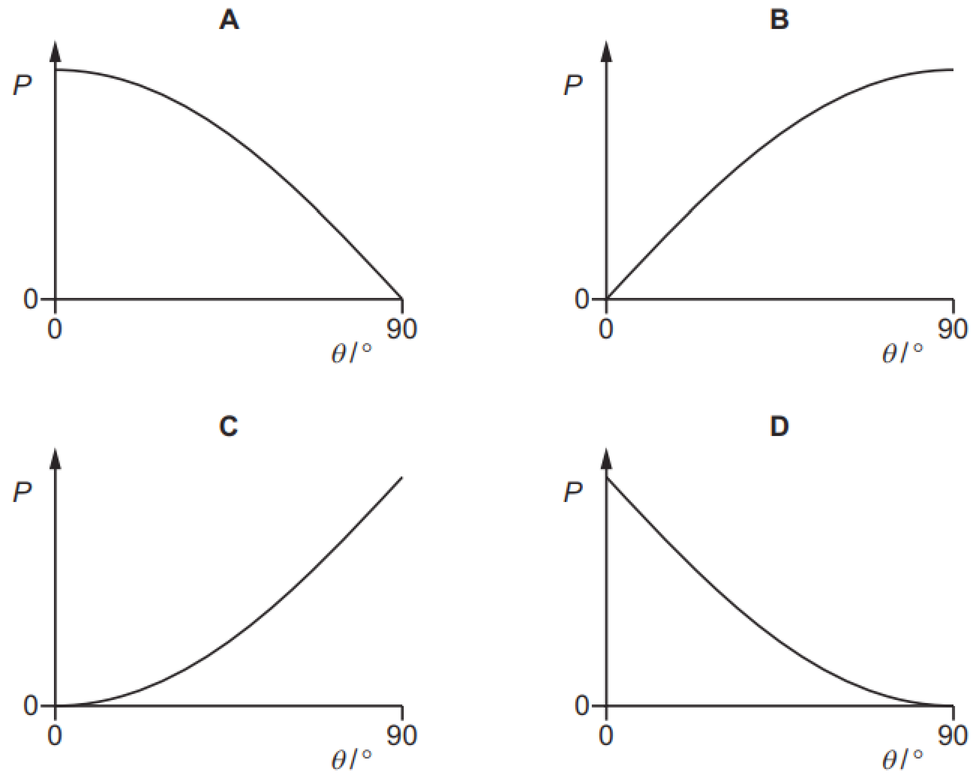
| | $p$ | $q$ |
|---|---|---|
| **A** | $-\frac{1}{2}$ | $-\frac{1}{2}$ |
| **B** | $-\frac{1}{2}$ | $\frac{1}{2}$ |
| **C** | $\frac{1}{2}$ | $-\frac{1}{2}$ |
| **D** | $\frac{1}{2}$ | $\frac{1}{2}$ |

**180. 9702/12/O/N/20/Q3**

The diagram shows a force $F$. $P$ is the horizontal component of $F$, at an angle $\theta$ to $F$.




Which graph best shows the variation with $\theta$ of the magnitude of $P$?




**181. 9702/13/O/N/20/Q1**

What is a reasonable estimate of the volume of a fully inflated standard football?

**A** 600 $cm^3$ **B** 6000 $cm^3$ **C** 60 000 $cm^3$ **D** 600 000 $cm^3$

**182. 9702/13/O/N/20/Q2**

What is **not** an SI base unit?

**A** coulomb

**B** kelvin

**C** kilogram

**D** second

**183. 9702/13/O/N/20/Q3**

Which two quantities are both vector quantities?

**A** displacement and distance

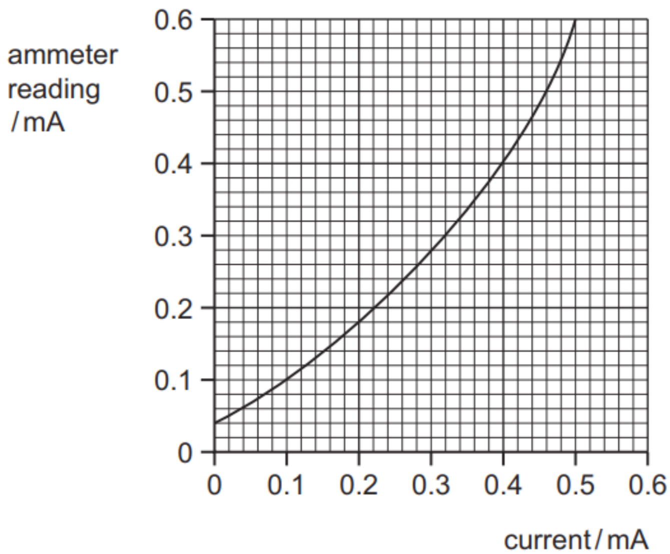
**B** force and momentum

**C** torque and time

**D** weight and pressure

**184. 9702/13/O/N/20/Q4**

A calibration curve is shown for an ammeter whose scale is inaccurate.




Two readings taken on the meter at different times during an experiment are 0.13 mA and 0.47 mA.

By how much did the current really increase between taking the two readings?

**A** 0.30 mA **B** 0.34 mA **C** 0.40 mA **D** 0.44 mA

**185. 9702/12/F/M/21/Q1**

What is a reasonable estimate for the density of sand?

**A** $2 \times 10^2\,\mathrm{g\,cm^{-3}}$

**B** $2 \times 10^3\,\mathrm{g\,cm^{-3}}$

**C** $2 \times 10^1\,\mathrm{kg\,m^{-3}}$

**D** $2 \times 10^3\,\mathrm{kg\,m^{-3}}$

**186. 9702/12/F/M/21/Q2**

Which physical quantity could have units of $\mathrm{N\,s^2\,m^{-1}}$?

**A** acceleration

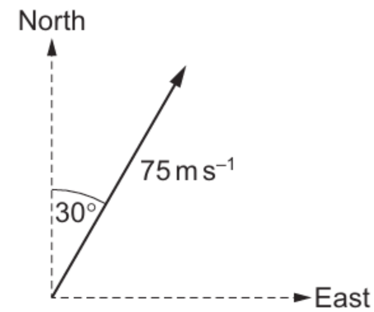
**B** force

**C** mass

**D** momentum

## 187. 9702/12/F/M/21/Q3

A velocity vector is shown.




What are the components of the velocity vector in the northerly and in the easterly directions?

| | component of vector in northerly direction / $m s^{-1}$ | component of vector in easterly direction / $m s^{-1}$ |
|---|---|---|
| **A** | 38 | 38 |
| **B** | 38 | 65 |
| **C** | 65 | 38 |
| **D** | 65 | 65 |

## 188. 9702/11/M/J/21/Q1

What is a reasonable estimate of the volume of an adult person?

**A** $0.10\,m^3$ **B** $0.50\,m^3$ **C** $1.0\,m^3$ **D** $2.0\,m^3$

## 189. 9702/11/M/J/21/Q2

Which combination of units could be used for expressing the power dissipated in a resistor?

**A** newton per second ($N\,s^{-1}$)

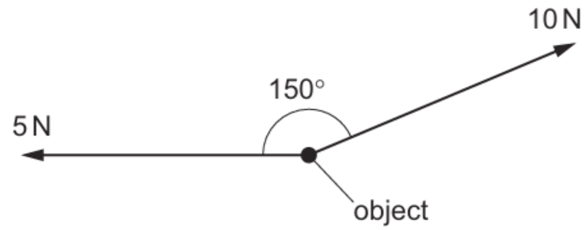
**B** newton second ($N\,s$)

**C** newton metre ($N\,m$)

**D** newton metre per second ($N\,m\,s^{-1}$)

**190. 9702/11/M/J/21/Q3**

A force of 10 N and a force of 5 N act on an object.




The angle between the forces is 150°.

The resultant force on the object can be resolved into a pair of perpendicular components.

Which row shows numerical expressions for a possible pair of perpendicular components?

| | force component / N | force component / N |
|---|---|---|
| **A** | $10\cos 30° - 5$ | $10\cos 30°$ |
| **B** | $10\sin 30° - 5$ | $10\cos 30°$ |
| **C** | $10 - 5\cos 30°$ | $5\sin 30°$ |
| **D** | $10 - 5\sin 30°$ | $5\cos 30°$ |

**191. 9702/11/M/J/21/Q5**

A micrometer screw gauge is used to measure the diameter of a wire.

The reading on the micrometer with the jaws closed is $(-0.05 \pm 0.02)$ mm.

The reading with the wire in position between the two jaws is $(+1.03 \pm 0.02)$ mm.

What is the diameter of the wire?

**A** $(0.98 \pm 0.02)$ mm

**B** $(1.08 \pm 0.02)$ mm

**C** $(0.98 \pm 0.04)$ mm

**D** $(1.08 \pm 0.04)$ mm

**192. 9702/12/M/J/21/Q1**

What is **not** a reasonable estimate of the physical property indicated?

**A** $2 \times 10^3$ W for the power dissipated by the heating element of an electric kettle

**B** $4 \times 10^2\,m^3$ for the volume of water in a swimming pool

**C** $5 \times 10^5$ N s for the momentum of a lorry moving along a road

**D** $6 \times 10^2$ N for the weight of a fully grown racehorse

**193. 9702/12/M/J/21/Q2**

Which quantity could have units of $N\,m\,V^{-1}$?

**A** acceleration

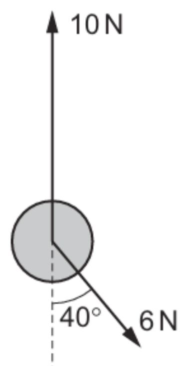
**B** charge

**C** current

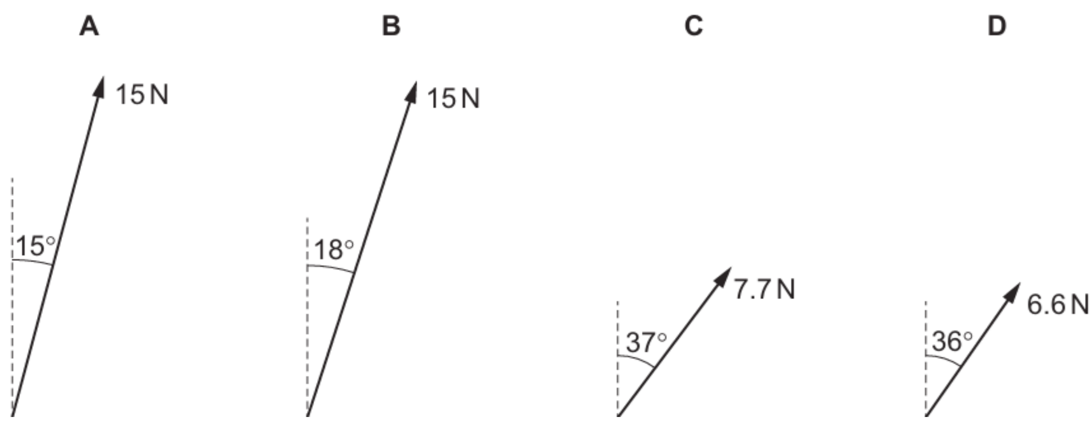
**D** resistance

**194. 9702/12/M/J/21/Q3**

An object is acted upon by two forces, 10 N in the vertical direction and 6 N at 40° to the vertical, as shown.




What is the resultant force acting on the object?




**195. 9702/13/M/J/21/Q1**

What is a reasonable estimate of the kinetic energy of an Olympic athlete sprinting in a 100 m race?

**A** 40 J **B** 400 J **C** 4000 J **D** 40 000 J

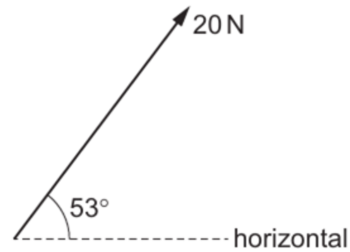
**196. 9702/13/M/J/21/Q2**

What is a unit of momentum?

**A** $kg\,m\,s^{-2}$ **B** $N\,s^{-1}$ **C** $N\,s$ **D** $kg\,s\,m^{-1}$

## 197. 9702/13/M/J/21/Q3

What is the horizontal component of the force shown?




**A** 12 N **B** 16 N **C** 25 N **D** 27 N

## 198. 9702/12/O/N/21/Q1

Which row shows what all physical quantities must have?

| | magnitude | direction | unit |
|---|---|---|---|
| **A** | ✓ | ✓ | ✓ |
| **B** | ✓ | ✓ | ✗ |
| **C** | ✓ | ✗ | ✓ |
| **D** | ✗ | ✗ | ✓ |

## 199. 9702/12/O/N/21/Q2

What is an alternative way of expressing an energy of 43 dJ?

**A** $4.3 \times 10^{3}$ mJ

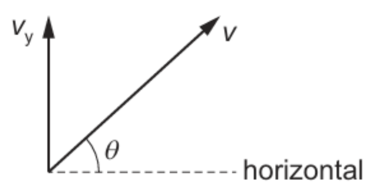
**B** $4.3 \times 10^{3}$ MJ

**C** $4.3 \times 10^{-3}$ mJ

**D** $4.3 \times 10^{-3}$ MJ

## 200. 9702/12/O/N/21/Q3

A tennis ball is hit so that it leaves the racket with velocity $v$ at an angle $\theta$ to the horizontal.




The vertical component of the velocity is $v_y$.

What is the magnitude of the horizontal component of $v$?

**A** $v\sin\theta$ **B** $v_y\cos\theta$ **C** $v_y\sin\theta$ **D** $(v^2 - v_y^{\,2})^{\frac{1}{2}}$

## 201. 9702/11/O/N/21/Q1

What is essential when recording a measurement of a physical quantity?

**A** the measurement has an SI unit

**B** the measurement has a unit and a number

**C** the measurement has a unit given as a base unit

**D** the measurement is from an analogue scale

## 202. 9702/11/O/N/21/Q2

The mobility $\mu$ of electrons travelling through a metal conductor can be calculated using the equation

$$\mu = \left(\frac{e}{m}\right)\tau$$

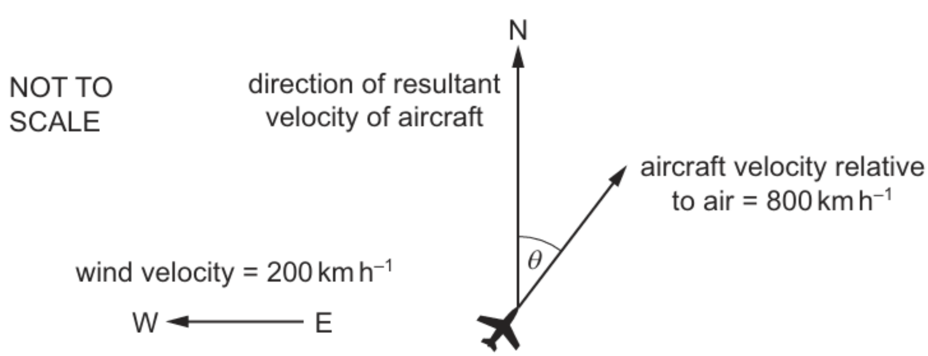
where $e$ is the charge on an electron and $m$ is its mass. The average time between the collisions of an electron with the atoms in the metal is $\tau$.

What are the SI base units of $\mu$?

**A** $\mathrm{A\,kg^{-1}}$ **B** $\mathrm{A\,s^2\,kg^{-1}}$ **C** $\mathrm{A\,s\,kg^{-1}}$ **D** $\mathrm{A\,s^{-2}\,kg^{-1}}$

## 203. 9702/11/O/N/21/Q3

An aircraft heads in a direction at an angle $\theta$ east of north with a horizontal velocity relative to the air of $800\,\mathrm{km\,h^{-1}}$. The wind blows with a horizontal velocity of $200\,\mathrm{km\,h^{-1}}$ from east to west, as shown.




The resultant velocity of the aircraft is in a direction due north.

What is angle $\theta$ and what is the magnitude of the resultant velocity?

| | $\theta/°$ | resultant velocity / $\mathrm{km\,h^{-1}}$ |
|---|---|---|
| **A** | 14 | 770 |
| **B** | 14 | 820 |
| **C** | 76 | 770 |
| **D** | 76 | 820 |

**204. 9702/13/O/N/21/Q2**

What is the unit of resistance when expressed in SI base units?

**A** $kg^{-1}\,m^{-2}\,s\,A^{2}$

**B** $kg^{-1}\,m^{-2}\,s^{3}\,A^{2}$

**C** $kg\,m^{2}\,s^{-1}\,A^{-2}$

**D** $kg\,m^{2}\,s^{-3}\,A^{-2}$

**205. 9702/13/O/N/21/Q3**

Which list consists only of scalar quantities?

**A** acceleration, displacement, force, weight

**B** density, energy, frequency, velocity

**C** distance, pressure, temperature, time

**D** momentum, power, volume, wavelength

**206. 9702/12/F/M/22/Q1**

What could **not** be a measurement of a physical quantity?

**A** 10 K **B** $11\,J\,N^{-1}\,m^{-1}$ **C** $17\,Pa\,m^{3}\,N^{-1}$ **D** 25 T m

**207. 9702/12/F/M/22/Q2**

A computer memory stick is labelled as having a storage capacity of 128 GB.

The letter B stands for byte, which is a unit.

What is the equivalent storage capacity?

**A** $1.28 \times 10^{8}$ B

**B** $1.28 \times 10^{11}$ B

**C** $1.28 \times 10^{14}$ B

**D** $1.28 \times 10^{17}$ B

**208. 9702/12/F/M/22/Q4**

Which statement about scalar and vector quantities is correct?

**A** A scalar quantity has direction but not magnitude.

**B** A scalar quantity has magnitude but not direction.

**C** A vector quantity has direction but not magnitude.

**D** A vector quantity has magnitude but not direction.

**209. 9702/11/M/J/22/Q1**

Which term represents a physical quantity?

**A** metre

**B** percentage uncertainty

**C** quark flavour

**D** spring constant

## 210. 9702/11/M/J/22/Q2

Which two units are identical when expressed in terms of SI base units?

**A** $\text{J}\,\text{C}^{-1}$ and $\text{kg}\,\text{m}^2\,\text{A}^{-1}\,\text{s}^{-2}$

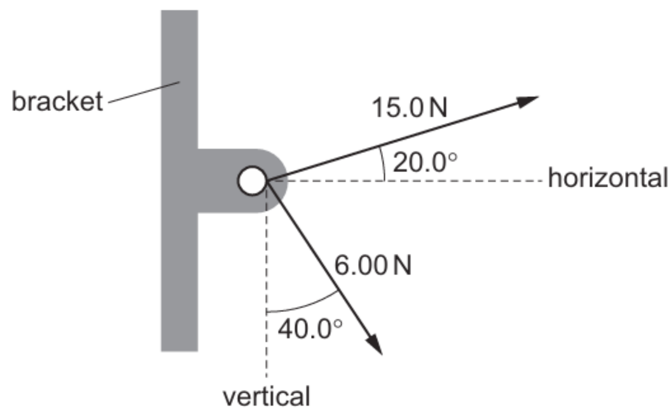
**B** $\text{J}\,\text{s}$ and $\text{kg}\,\text{m}^2\,\text{s}^{-1}$

**C** $\text{N}\,\text{m}$ and $\text{kg}\,\text{m}^3\,\text{s}^{-2}$

**D** $\text{N}\,\text{s}$ and $\text{kg}\,\text{m}\,\text{s}^{-3}$

## 211. 9702/11/M/J/22/Q4

Two cables are attached to a bracket and exert forces as shown.




What are the magnitudes of the horizontal and vertical components of the resultant of the two forces?

| | horizontal component / N | vertical component / N |
|---|---|---|
| **A** | 9.73 | 0.534 |
| **B** | 9.73 | 10.2 |
| **C** | 18.0 | 0.534 |
| **D** | 18.0 | 10.2 |

## 212. 9702/12/M/J/22/Q1

Which estimate is reasonable?

**A** $1 \times 10^{-3}$ kg for the mass of a grain of sand

**B** $1 \times 10^{-2}\,\text{m}^3$ for the volume of a tennis ball

**C** $1 \times 10^{0}$ J for the work done lifting an apple from waist height to head height

**D** $1 \times 10^{4}$ W for the power of a light bulb in a house

## 213. 9702/12/M/J/22/Q2

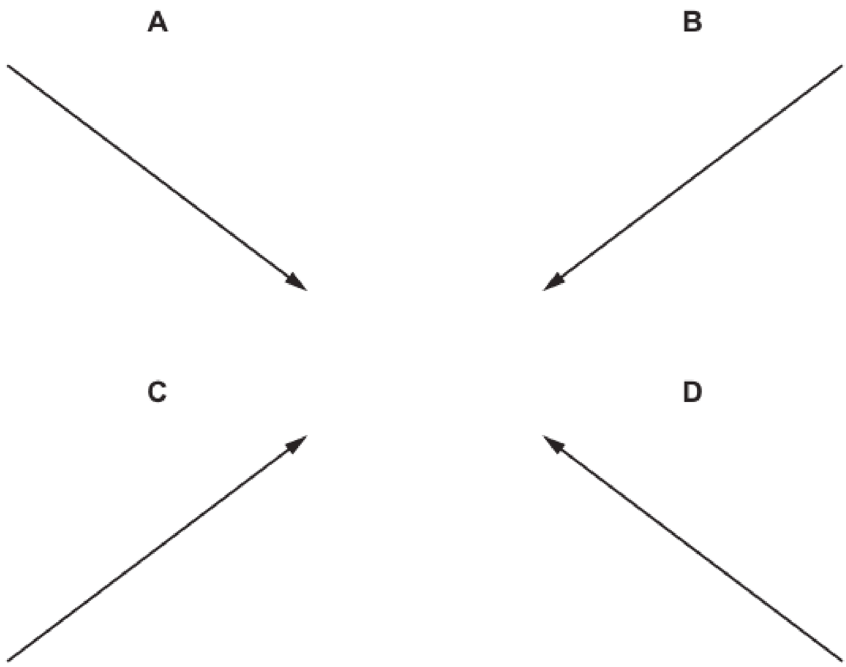
What is the symbol for the SI base unit of temperature?

**A** C **B** K **C** °C **D** °K

**214. 9702/12/M/J/22/Q4**

An object is moving with an initial velocity of $4.0\,m\,s^{-1}$ to the right. The velocity of the object changes so that its final velocity is $3.0\,m\,s^{-1}$ downwards, as shown.


Which arrow represents the change in velocity of the object?




**215. 9702/13/M/J/22/Q1**

Which pair of quantities are physical quantities?

**A** charge and ampere

**B** efficiency and kilogram

**C** pascal and strain

**D** period and potential difference

**216. 9702/13/M/J/22/Q2**

Which list of unit prefixes decreases in magnitude from left to right?

**A** centi, deci, milli

**B** deci, milli, centi

**C** pico, kilo, milli

**D** kilo, milli, pico

**217. 9702/13/M/J/22/Q3**

The drag coefficient $C_d$ is a number with no units. It is used to compare the drag on different cars at different speeds. $C_d$ is given by the equation

$$C_d = \frac{2F}{v^n \rho A}$$

where $F$ is the drag force on the car, $\rho$ is the density of the air, $A$ is the cross-sectional area of the car and $v$ is the speed of the car.

What is the value of $n$?

**A** 1 **B** 2 **C** 3 **D** 4

**218. 9702/13/M/J/22/Q5**

Forces of magnitudes 2 N, 4 N and 7 N combine to produce a resultant force.

The magnitudes of the three forces are fixed, but the forces may act in any direction in the same plane.

What is **not** a possible magnitude of the resultant force?

**A** 0 N **B** 5 N **C** 8 N **D** 13 N

**219. 9702/11/O/N/22/Q1**

What is needed to accurately represent all physical quantities?

**A** a base unit and a number

**B** a unit and a number expressed in standard form (scientific notation)

**C** a unit and a numerical magnitude

**D** an SI unit and a numerical magnitude

**220. 9702/11/O/N/22/Q2**

A voltmeter connected across a resistor in a circuit reads 3.6 V.

What could be the current in the resistor and the resistance of the resistor?

| | current | resistance |
|---|---|---|
| **A** | 150 mA | 0.24 kΩ |
| **B** | 15 mA | 2.4 kΩ |
| **C** | 1.5 mA | 0.24 MΩ |
| **D** | 15 μA | 240 kΩ |

**221. 9702/11/O/N/22/Q4**

Which quantity is a vector?

**A** momentum

**B** speed

**C** temperature

**D** Young modulus

**222. 9702/12/O/N/22/Q1**

Which quantity is a physical quantity?

**A** flavour

**B** kelvin

**C** minute

**D** potential difference

**223. 9702/12/O/N/22/Q2**

What is a power of 3.7 MW when expressed in kilowatts?

**A** $3.7 \times 10^{-3}$ kW

**B** $3.7 \times 10^{-3}$ KW

**C** $3.7 \times 10^{3}$ kW

**D** $3.7 \times 10^{3}$ KW

**224. 9702/12/O/N/22/Q4**

What is the difference between a scalar quantity and a vector quantity?

**A** A scalar quantity has direction but a vector quantity does not.

**B** A scalar quantity has magnitude but a vector quantity does not.

**C** A vector quantity has direction but a scalar quantity does not.

**D** A vector quantity has magnitude but a scalar quantity does not.

**225. 9702/13/O/N/22/Q1**

A train of mass 600 000 kg moves with a speed of 100 km $h^{-1}$.

What is the order of magnitude of the kinetic energy of the train?

**A** $10^{6}$ J **B** $10^{8}$ J **C** $10^{10}$ J **D** $10^{12}$ J

**226. 9702/13/O/N/22/O2**

What are the SI base units of electromotive force (e.m.f.)?

**A** kg $m^{2} s^{-1} A^{-1}$

**B** kg $m^{2} s^{-3} A^{-1}$

**C** kg $m^{2} s^{-1} A$

**D** kg m $s^{-3} A^{-1}$

## 227. 9702/13/O/N/22/Q4

Which quantity is a vector quantity?

**A** density

**B** mass

**C** volume

**D** weight

## 228. 9702/12/F/M/23/Q1

What represents a physical quantity?

**A** 3.0

**B** kilogram

**C** 7.0 N

**D** 40%

## 229. 9702/12/F/M/23/Q2

The relationship between the variables *D* and *T* is given by the equation

$$\frac{1}{T} = \frac{b}{\sqrt{D}} + c$$

where *b* and *c* are constants.

The unit of *D* is $m^2$ and the unit of *T* is s.

What are the units of *b* and *c*?

| | unit of *b* | unit of *c* |
|---|---|---|
| **A** | m s | s |
| **B** | $m\,s^{-1}$ | $s^{-1}$ |
| **C** | $m^{-1}\,s$ | s |
| **D** | $m^{-1}\,s^{-1}$ | $s^{-1}$ |

## 230. 9702/11/M/J/23/Q1

Which unit is **not** an SI base unit?

**A** A **B** kg **C** C **D** s

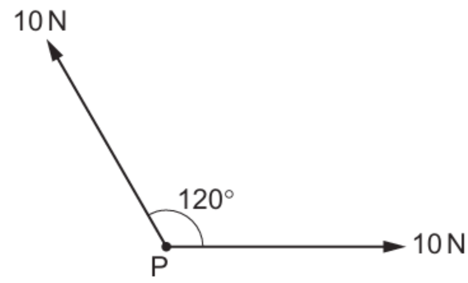
## 231. 9702/11/M/J/23/Q2

What is the best estimate of the number of atoms in a piece of metal of volume 50 $cm^3$?

**A** $5 \times 10^{15}$ **B** $5 \times 10^{25}$ **C** $5 \times 10^{29}$ **D** $5 \times 10^{31}$

**232. 9702/11/M/J/23/Q4**

Two forces, each of 10 N, act at a point P, as shown. The angle between the directions of the forces is 120°.




What is the magnitude of the resultant force?

**A** 5 N **B** 10 N **C** 17 N **D** 20 N

**233. 9702/12/M/J/23/Q1**

A stone sinks in water.

What is a possible value for the density of the stone?

**A** $8 \times 10^2\,\mathrm{kg\,m^{-3}}$

**B** $2 \times 10^3\,\mathrm{kg\,m^{-3}}$

**C** $8 \times 10^3\,\mathrm{N\,m^{-3}}$

**D** $2 \times 10^4\,\mathrm{N\,m^{-3}}$

**234. 9702/12/M/J/23/Q2**

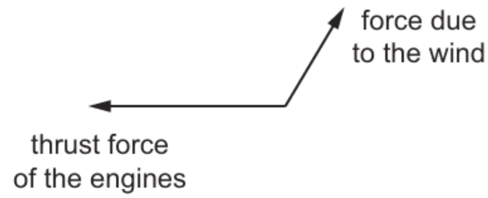
Gm, Tm, µm and pm are all units of length.

Which unit is the largest and which unit is the smallest?

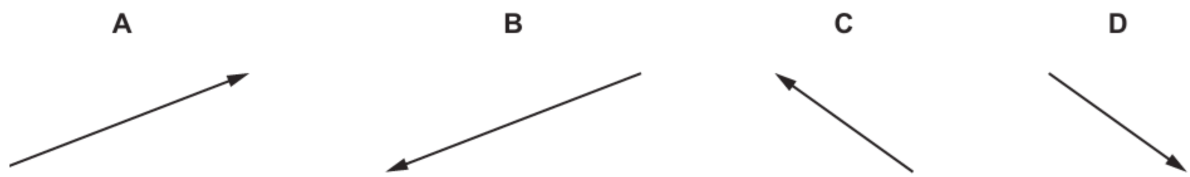
| | largest unit | smallest unit |
|---|---|---|
| **A** | Gm | µm |
| **B** | Gm | pm |
| **C** | Tm | µm |
| **D** | Tm | pm |

**235. 9702/12/M/J/23/Q4**

An aircraft travels along a horizontal path. Two of the forces that act horizontally on the aircraft are the thrust force of the engines and the force due to the wind. The vector diagram for these forces is shown.




Which vector represents the resultant horizontal force acting on the aircraft due to these two forces?




**236. 9702/13/M/J/23/Q1**

What **must** be included in a record of a physical quantity?

**A** an integer value for the quantity

**B** an SI unit

**C** a numerical value for the quantity

**D** a unit expressed in base units

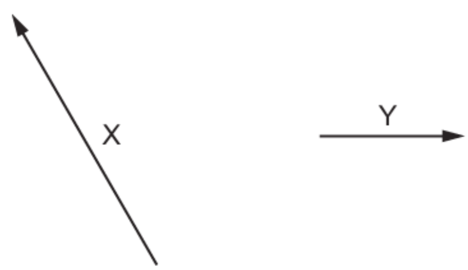
**237. 9702/13/M/J/23/Q2**

What is the ohm expressed in SI base units?

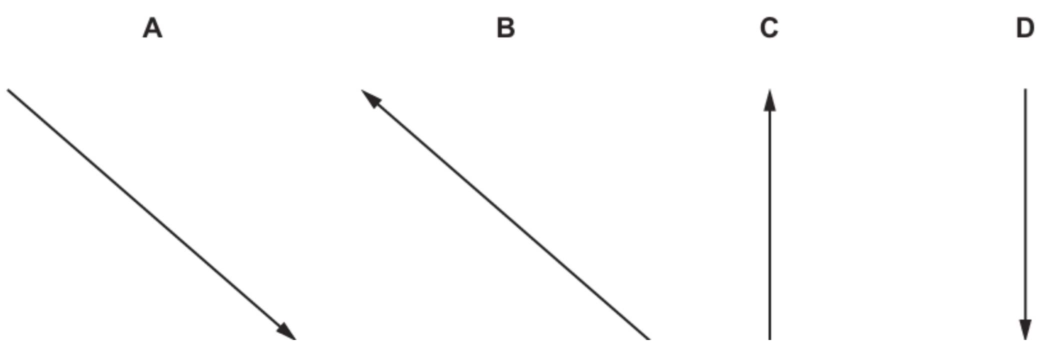
**A** $kg\,m^2\,s^{-3}\,A^{-2}$ **B** $kg^{-1}\,m^{-2}\,s^3\,A^2$ **C** $J\,C^{-1}\,A^{-1}$ **D** $W\,A^{-2}$

**238. 9702/13/M/J/23/Q4**

The diagram shows two vectors, X and Y, drawn to scale.




If X = Y – Z, which diagram represents the vector Z?



**239. 9702/11/O/N/23/Q1**

What is a reasonable estimate of the cross-sectional area of the wire in a paper clip?

**A** $1 \times 10^{-3}\,m^2$ **B** $8 \times 10^{-5}\,m^2$ **C** $8 \times 10^{-7}\,m^2$ **D** $1 \times 10^{-9}\,m^2$

**240. 9702/11/O/N/23/Q2**

Which quantity is **not** an SI base quantity?

**A** charge

**B** mass

**C** temperature

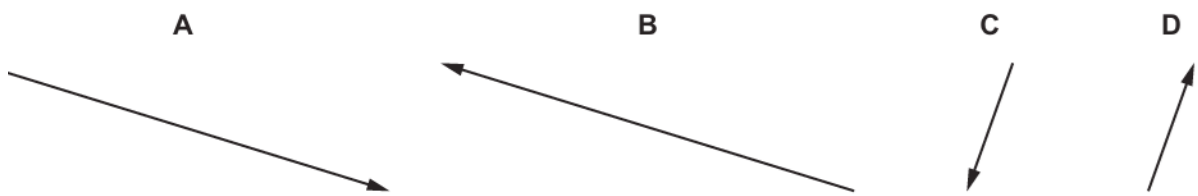
**D** time

**241. 9702/11/O/N/23/Q4**

The diagram shows two coplanar forces, $P$ and $Q$, drawn to scale.


Force $R$ is given by $R = Q - P$.

Which diagram represents $R$?




**242. 9702/12/O/N/23/Q1**

A student estimates the maximum speed of some different moving objects.

Which maximum speed is **not** a reasonable estimate?

**A** container ship: $10\,m\,s^{-1}$

**B** Olympic sprinter: $0.1\,km\,s^{-1}$

**C** racing car: $9000\,cm\,s^{-1}$

**D** snail: $0.01\,km\,h^{-1}$

**243. 9702/12/O/N/23/Q2**

Which quantity is an SI base quantity?

**A** force

**B** newton

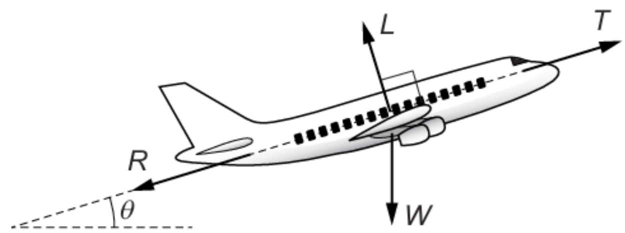
**C** second

**D** time

**244. 9702/12/O/N/23/Q4**

An aeroplane is moving at a constant speed in a straight line at an angle $\theta$ to the horizontal.

Four forces act on the aeroplane: thrust force $T$, weight $W$, lift force $L$ and resistive force $R$.




Which two equations must be correct?

**A** $L = W\cos\theta$ and $T = R + W\sin\theta$

**B** $L = W\sin\theta$ and $T = R + W\cos\theta$

**C** $L = W\cos\theta$ and $T = R - W\sin\theta$

**D** $L = W\sin\theta$ and $T = R - W\cos\theta$

**245. 9702/13/O/N/23/Q1**

What is the best estimate of the wavelength of green light?

**A** 260 nm **B** 540 nm **C** 780 nm **D** 920 nm

**246. 9702/13/O/N/23/Q2**

In an electric circuit, an ammeter reads 2 μA.

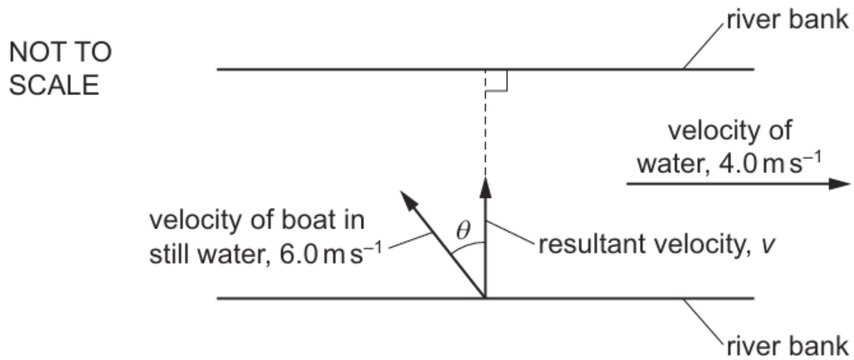
In a second circuit, the ammeter reads 1 mA.

How many times larger is the current in the second circuit compared with the current in the first circuit?

**A** 500 **B** 5000 **C** 500 000 **D** 5 000 000

**247. 9702/13/O/N/23/Q4**

A boat is crossing a river in which the water is moving at a speed of $4.0\,m\,s^{-1}$ from left to right.




In still water, the speed of the boat is $6.0\,m\,s^{-1}$. The boat is directed at an angle $\theta$ to a line perpendicular to the river banks. The resultant velocity $v$ of the boat is in a direction perpendicular to the river banks.

What are the values of $\theta$ and $v$?

| | $\theta$ / ° | $v$ / $m\,s^{-1}$ |
|---|---|---|
| **A** | 42 | 4.5 |
| **B** | 42 | 7.2 |
| **C** | 48 | 4.5 |
| **D** | 48 | 7.2 |